

Customer Segmentation and Shopping Pattern Analysis

Mini Project - Unsupervised Learning

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Project Overview

This project analyzes consumer behavior and shopping habits through customer segmentation using multiple clustering algorithms and identifies key shopping patterns that influence purchasing decisions.

1. Data Preprocessing (2 marks)

Tasks:

- Load and clean the dataset
- Handle missing values and duplicates
- Perform outlier detection and removal
- Feature scaling/normalization
- Create relevant features for both clustering and association analysis

```
In [1]: # Import necessary libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.cluster import KMeans, AgglomerativeClustering, DBSCAN
from sklearn.metrics import silhouette_score, silhouette_samples
from scipy.cluster.hierarchy import dendrogram, linkage
import warnings
warnings.filterwarnings('ignore')
```

```
# Set style for visualizations
sns.set_style("whitegrid")
plt.rcParams['figure.figsize'] = (12, 6)

# Load the dataset
df = pd.read_csv('Shopping Trends And Customer Behaviour Dataset.csv')
print("Dataset loaded successfully!")
print(f"Dataset shape: {df.shape}")
print("\nFirst few rows:")
print(df.head())
```

Dataset loaded successfully!

Dataset shape: (3900, 18)

First few rows:

	Unnamed: 0	Customer ID	Age	Gender	Item Purchased	Category	\
0	0	1	55	Male	Blouse	Clothing	
1	1	2	19	Male	Sweater	Clothing	
2	2	3	50	Male	Jeans	Clothing	
3	3	4	21	Male	Sandals	Footwear	
4	4	5	45	Male	Blouse	Clothing	

	Purchase Amount (USD)	Location	Color	Season	Review Rating	\
0	53	Kentucky	Gray	Winter	3.1	
1	64	Maine	Maroon	Winter	3.1	
2	73	Massachusetts	Maroon	Spring	3.1	
3	90	Rhode Island	Maroon	Spring	3.5	
4	49	Oregon	Turquoise	Spring	2.7	

	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	\
0	Yes	Express	Yes	Yes	
1	Yes	Express	Yes	Yes	
2	Yes	Free Shipping	Yes	Yes	
3	Yes	Next Day Air	Yes	Yes	
4	Yes	Free Shipping	Yes	Yes	

	Previous Purchases	Payment Method	Frequency of Purchases
0	14	Venmo	Fortnightly
1	2	Cash	Fortnightly
2	23	Credit Card	Weekly
3	49	PayPal	Weekly
4	31	PayPal	Annually

```
In [2]: # Check for missing values
print("Missing values in dataset:")
print(df.isnull().sum())
```

```
print(f"\nTotal missing values: {df.isnull().sum().sum()}")

# Check for duplicates
print(f"\nDuplicate rows: {df.duplicated().sum()}")

# Display basic statistics
print("\nDataset Info:")
print(df.info())
print("\nBasic Statistics:")
print(df.describe())
```

Missing values in dataset:

Unnamed: 0	0
Customer ID	0
Age	0
Gender	0
Item Purchased	0
Category	0
Purchase Amount (USD)	0
Location	0
Color	0
Season	0
Review Rating	0
Subscription Status	0
Shipping Type	0
Discount Applied	0
Promo Code Used	0
Previous Purchases	0
Payment Method	0
Frequency of Purchases	0

dtype: int64

Total missing values: 0

Duplicate rows: 0

Dataset Info:

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 3900 entries, 0 to 3899

Data columns (total 18 columns):

#	Column	Non-Null Count	Dtype
----	-----	-----	-----
0	Unnamed: 0	3900 non-null	int64
1	Customer ID	3900 non-null	int64
2	Age	3900 non-null	int64
3	Gender	3900 non-null	object
4	Item Purchased	3900 non-null	object
5	Category	3900 non-null	object
6	Purchase Amount (USD)	3900 non-null	int64
7	Location	3900 non-null	object
8	Color	3900 non-null	object
9	Season	3900 non-null	object
10	Review Rating	3900 non-null	float64
11	Subscription Status	3900 non-null	object
12	Shipping Type	3900 non-null	object
13	Discount Applied	3900 non-null	object
14	Promo Code Used	3900 non-null	object

```

15 Previous Purchases      3900 non-null    int64
16 Payment Method          3900 non-null    object
17 Frequency of Purchases  3900 non-null    object
dtypes: float64(1), int64(5), object(12)
memory usage: 548.6+ KB
None

```

Basic Statistics:

	Unnamed: 0	Customer ID	Age	Purchase Amount (USD)	\
count	3900.000000	3900.000000	3900.000000	3900.000000	
mean	1949.500000	1950.500000	44.068462	59.764359	
std	1125.977353	1125.977353	15.207589	23.685392	
min	0.000000	1.000000	18.000000	20.000000	
25%	974.750000	975.750000	31.000000	39.000000	
50%	1949.500000	1950.500000	44.000000	60.000000	
75%	2924.250000	2925.250000	57.000000	81.000000	
max	3899.000000	3900.000000	70.000000	100.000000	

	Review Rating	Previous Purchases
count	3900.000000	3900.000000
mean	3.749949	25.351538
std	0.716223	14.447125
min	2.500000	1.000000
25%	3.100000	13.000000
50%	3.700000	25.000000
75%	4.400000	38.000000
max	5.000000	50.000000

```

In [3]: # Data Cleaning and Preprocessing
df_clean = df.copy()

# Remove duplicates
df_clean = df_clean.drop_duplicates()
print(f"Duplicates removed. Shape after removing duplicates: {df_clean.shape}")

# Handle the index column if it exists
if df_clean.columns[0] == 'Unnamed: 0':
    df_clean = df_clean.drop('Unnamed: 0', axis=1)

# Identify numeric columns
numeric_cols = df_clean.select_dtypes(include=[np.number]).columns.tolist()
print(f"\nNumeric columns: {numeric_cols}")

# Outlier detection using IQR method for numeric columns
def detect_and_handle_outliers(data, cols):
    Q1 = data[cols].quantile(0.25)

```

```

Q3 = data[cols].quantile(0.75)
IQR = Q3 - Q1
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

outliers = ((data[cols] < lower_bound) | (data[cols] > upper_bound)).any(axis=1)
print(f"Outliers detected: {outliers.sum()} rows")
return data[~outliers]

df_clean = detect_and_handle_outliers(df_clean, numeric_cols)
print(f"Dataset shape after outlier removal: {df_clean.shape}")
print("\nData cleaned successfully!")

```

Duplicates removed. Shape after removing duplicates: (3900, 18)

Numeric columns: ['Customer ID', 'Age', 'Purchase Amount (USD)', 'Review Rating', 'Previous Purchases']

Outliers detected: 0 rows

Dataset shape after outlier removal: (3900, 17)

Data cleaned successfully!

```

In [4]: # Feature Engineering for clustering
# Encode categorical variables
le_dict = {}
categorical_cols = df_clean.select_dtypes(include=['object']).columns

df_encoded = df_clean.copy()

for col in categorical_cols:
    le = LabelEncoder()
    df_encoded[col] = le.fit_transform(df_encoded[col])
    le_dict[col] = le
    print(f"Encoded {col}: {len(le.classes_)} unique values")

print(f"\nDataset shape after encoding: {df_encoded.shape}")
print("\nEncoded dataset sample:")
print(df_encoded.head())

```

Encoded Gender: 2 unique values
 Encoded Item Purchased: 25 unique values
 Encoded Category: 4 unique values
 Encoded Location: 50 unique values
 Encoded Color: 25 unique values
 Encoded Season: 4 unique values
 Encoded Subscription Status: 2 unique values
 Encoded Shipping Type: 6 unique values
 Encoded Discount Applied: 2 unique values
 Encoded Promo Code Used: 2 unique values
 Encoded Payment Method: 6 unique values
 Encoded Frequency of Purchases: 7 unique values

Dataset shape after encoding: (3900, 17)

Encoded dataset sample:

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	\
0	1	55	1	2	1	53	
1	2	19	1	23	1	64	
2	3	50	1	11	1	73	
3	4	21	1	14	2	90	
4	5	45	1	2	1	49	

	Location	Color	Season	Review Rating	Subscription Status	Shipping Type	\
0	16	7	3	3.1	1	1	
1	18	12	3	3.1	1	1	
2	20	12	1	3.1	1	2	
3	38	12	1	3.5	1	3	
4	36	21	1	2.7	1	2	

	Discount Applied	Promo Code Used	Previous Purchases	Payment Method	\
0	1	1	14	5	
1	1	1	2	1	
2	1	1	23	2	
3	1	1	49	4	
4	1	1	31	4	

	Frequency of Purchases
0	3
1	3
2	6
3	6
4	0

```
In [5]: # Feature Scaling/Normalization using StandardScaler
scaler = StandardScaler()
```

```
df_scaled = scaler.fit_transform(df_encoded)
df_scaled = pd.DataFrame(df_scaled, columns=df_encoded.columns)

print("Feature scaling completed!")
print(f"\nScaled data shape: {df_scaled.shape}")
print("\nScaled data statistics (should be centered around 0):")
print(df_scaled.describe())

# Prepare data for clustering
X_cluster = df_scaled.copy()
print(f"\nData prepared for clustering: {X_cluster.shape}")
```


Feature scaling completed!

Scaled data shape: (3900, 17)

Scaled data statistics (should be centered around 0):

	Customer ID	Age	Gender	Item Purchased	Category \
count	3.900000e+03	3.900000e+03	3900.000000	3.900000e+03	3.900000e+03
mean	5.830094e-17	-1.985876e-16	0.000000	6.923237e-17	1.667043e-16
std	1.000128e+00	1.000128e+00	1.000128	1.000128e+00	1.000128e+00
min	-1.731607e+00	-1.714394e+00	-1.457738	-1.671990e+00	-1.117360e+00
25%	-8.658034e-01	-8.594483e-01	-1.457738	-8.384529e-01	-1.117360e+00
50%	0.000000e+00	-4.502378e-03	0.685994	-4.915732e-03	-2.001925e-03
75%	8.658034e-01	8.504436e-01	0.685994	8.286215e-01	-2.001925e-03
max	1.731607e+00	1.705390e+00	0.685994	1.662159e+00	2.228715e+00

	Purchase Amount (USD)	Location	Color	Season \
count	3.900000e+03	3.900000e+03	3.900000e+03	3.900000e+03
mean	-1.166019e-16	-4.372571e-17	-1.293552e-16	8.380760e-17
std	1.000128e+00	1.000128e+00	1.000128e+00	1.000128e+00
min	-1.679071e+00	-1.692046e+00	-1.677060e+00	-1.336344e+00
25%	-8.767861e-01	-8.553111e-01	-8.461114e-01	-6.649584e-01
50%	9.950067e-03	-1.857623e-02	-1.516303e-02	-4.411633e-01
75%	8.966862e-01	8.181586e-01	8.157853e-01	4.540172e-01
max	1.698971e+00	1.724621e+00	1.646734e+00	1.349198e+00

	Review Rating	Subscription Status	Shipping Type	Discount Applied \
count	3.900000e+03	3.900000e+03	3.900000e+03	3900.000000
mean	4.463666e-16	5.830094e-17	-1.056705e-16	0.000000
std	1.000128e+00	1.000128e+00	1.000128e+00	1.000128
min	-1.745419e+00	-6.081636e-01	-1.481324e+00	-0.868554
25%	-9.075836e-01	-6.081636e-01	-8.921780e-01	-0.868554
50%	-6.974802e-02	-6.081636e-01	2.861133e-01	-0.868554
75%	9.077268e-01	1.644294e+00	8.752589e-01	1.151339
max	1.745562e+00	1.644294e+00	1.464405e+00	1.151339

	Promo Code Used	Previous Purchases	Payment Method \
count	3900.000000	3.900000e+03	3.900000e+03
mean	0.000000	7.287618e-17	-2.368476e-17
std	1.000128	1.000128e+00	1.000128e+00
min	-0.868554	-1.685779e+00	-1.486198e+00
25%	-0.868554	-8.550575e-01	-8.946310e-01
50%	-0.868554	-2.433588e-02	-3.030642e-01
75%	1.151339	8.756125e-01	8.800693e-01
max	1.151339	1.706334e+00	1.471636e+00

Frequency of Purchases

count	3.900000e+03
mean	-2.550666e-17
std	1.000128e+00
min	-1.488699e+00
25%	-9.882745e-01
50%	1.257477e-02
75%	1.013424e+00
max	1.513849e+00

Data prepared for clustering: (3900, 17)

2. Customer Segmentation using Clustering (5 marks)

2.1 K-Means Clustering with Elbow Method (1.5 marks)

Elbow Method to find optimal number of clusters

```
In [6]: # Elbow Method for K-means
inertias = []
silhouette_scores = []
K_range = range(2, 11)

for k in K_range:
    kmeans = KMeans(n_clusters=k, random_state=42, n_init=10)
    kmeans.fit(X_cluster)
    inertias.append(kmeans.inertia_)
    sil_score = silhouette_score(X_cluster, kmeans.labels_)
    silhouette_scores.append(sil_score)

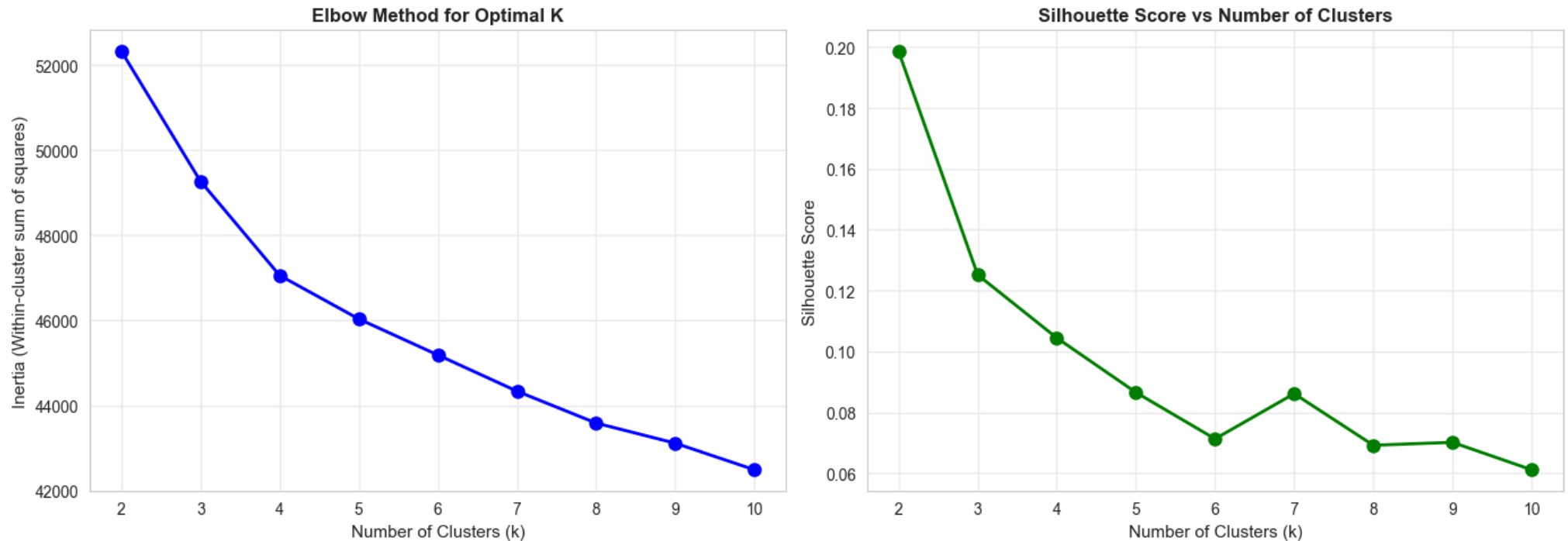
# Plot Elbow curve
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

axes[0].plot(K_range, inertias, 'bo-', linewidth=2, markersize=8)
axes[0].set_xlabel('Number of Clusters (k)', fontsize=11)
axes[0].set_ylabel('Inertia (Within-cluster sum of squares)', fontsize=11)
axes[0].set_title('Elbow Method for Optimal K', fontsize=12, fontweight='bold')
axes[0].grid(True, alpha=0.3)

axes[1].plot(K_range, silhouette_scores, 'go-', linewidth=2, markersize=8)
axes[1].set_xlabel('Number of Clusters (k)', fontsize=11)
axes[1].set_ylabel('Silhouette Score', fontsize=11)
axes[1].set_title('Silhouette Score vs Number of Clusters', fontsize=12, fontweight='bold')
axes[1].grid(True, alpha=0.3)
```

```
plt.tight_layout()
plt.show()
```

```
optimal_k = K_range[np.argmax(silhouette_scores)]
print(f"Optimal number of clusters based on Elbow method and Silhouette: {optimal_k}")
print(f"Silhouette scores for each k: {dict(zip(K_range, [f'{s:.4f}' for s in silhouette_scores]))}")
```



Optimal number of clusters based on Elbow method and Silhouette: 2

Silhouette scores for each k: {2: '0.1987', 3: '0.1252', 4: '0.1044', 5: '0.0865', 6: '0.0713', 7: '0.0861', 8: '0.0692', 9: '0.0701', 10: '0.0611'}

```
In [7]: # Apply K-means with optimal k
kmeans_final = KMeans(n_clusters=optimal_k, random_state=42, n_init=10)
kmeans_clusters = kmeans_final.fit_predict(X_cluster)
kmeans_silhouette = silhouette_score(X_cluster, kmeans_clusters)

print(f"\nK-Means Clustering Results (k={optimal_k}):")
print(f"Silhouette Score: {kmeans_silhouette:.4f}")
print(f"Cluster distribution:\n{pd.Series(kmeans_clusters).value_counts().sort_index()}")

# Visualize K-means clusters using PCA for 2D representation
from sklearn.decomposition import PCA

pca = PCA(n_components=2)
```

```

X_pca = pca.fit_transform(X_cluster)

plt.figure(figsize=(12, 5))

plt.subplot(1, 2, 1)
scatter = plt.scatter(X_pca[:, 0], X_pca[:, 1], c=kmeans_clusters, cmap='viridis', s=50, alpha=0.6)
plt.scatter(pca.transform(kmeans_final.cluster_centers_)[:, 0],
            pca.transform(kmeans_final.cluster_centers_)[:, 1],
            marker='X', s=300, c='red', edgecolors='black', linewidth=2)
plt.xlabel(f'PC1 ({pca.explained_variance_ratio_[0]:.2%})', fontsize=11)
plt.ylabel(f'PC2 ({pca.explained_variance_ratio_[1]:.2%})', fontsize=11)
plt.title('K-Means Clustering (PCA Visualization)', fontsize=12, fontweight='bold')
plt.colorbar(scatter, label='Cluster')
plt.grid(True, alpha=0.3)

# Silhouette plot for K-means
plt.subplot(1, 2, 2)
silhouette_vals = silhouette_samples(X_cluster, kmeans_clusters)
y_lower = 10
colors = plt.cm.viridis(np.linspace(0, 1, optimal_k))

for i in range(optimal_k):
    cluster_silhouette_vals = silhouette_vals[kmeans_clusters == i]
    cluster_silhouette_vals.sort()
    size_cluster_i = cluster_silhouette_vals.shape[0]
    y_upper = y_lower + size_cluster_i
    plt.fill_betweenx(np.arange(y_lower, y_upper), 0, cluster_silhouette_vals,
                     facecolor=colors[i], edgecolor=colors[i], alpha=0.7)
    y_lower = y_upper + 10

plt.xlabel('Silhouette Coefficient', fontsize=11)
plt.ylabel('Cluster Label', fontsize=11)
plt.title('Silhouette Plot for K-Means Clusters', fontsize=12, fontweight='bold')
plt.axvline(x=kmeans_silhouette, color='red', linestyle='--', label=f'Average: {kmeans_silhouette:.3f}')
plt.legend()
plt.grid(True, alpha=0.3)

plt.tight_layout()
plt.show()

```

K-Means Clustering Results (k=2):

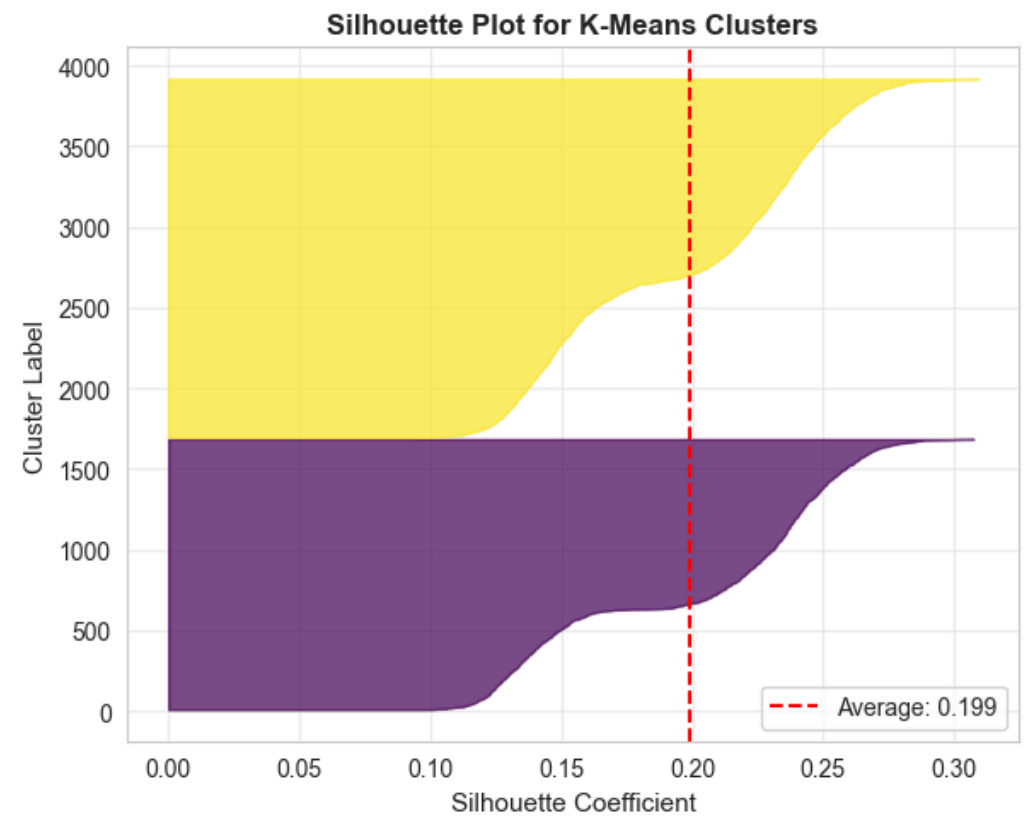
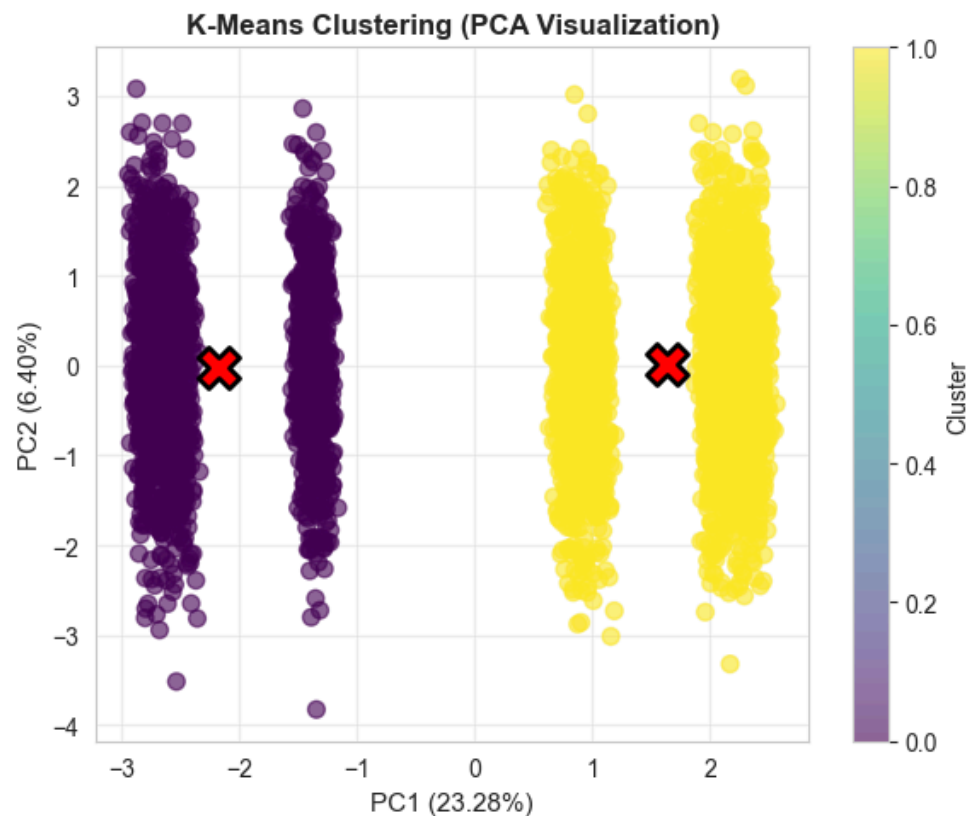
Silhouette Score: 0.1987

Cluster distribution:

0 1677

1 2223

Name: count, dtype: int64



2.2 Hierarchical Clustering (1.5 marks)

```
In [8]: # Test different linkage methods
linkage_methods = ['ward', 'complete', 'average', 'single']
hierarchical_silhouettes = {}

for method in linkage_methods:
    hc = AgglomerativeClustering(n_clusters=optimal_k, linkage=method)
    hc_clusters = hc.fit_predict(X_cluster)
    sil_score = silhouette_score(X_cluster, hc_clusters)
    hierarchical_silhouettes[method] = sil_score
    print(f"Hierarchical ({method}) - Silhouette Score: {sil_score:.4f}")

# Use best linkage method
best_linkage = max(hierarchical_silhouettes, key=hierarchical_silhouettes.get)
print(f"\nBest linkage method: {best_linkage}")

# Create dendrogram with best method
Z = linkage(X_cluster, method=best_linkage)
```

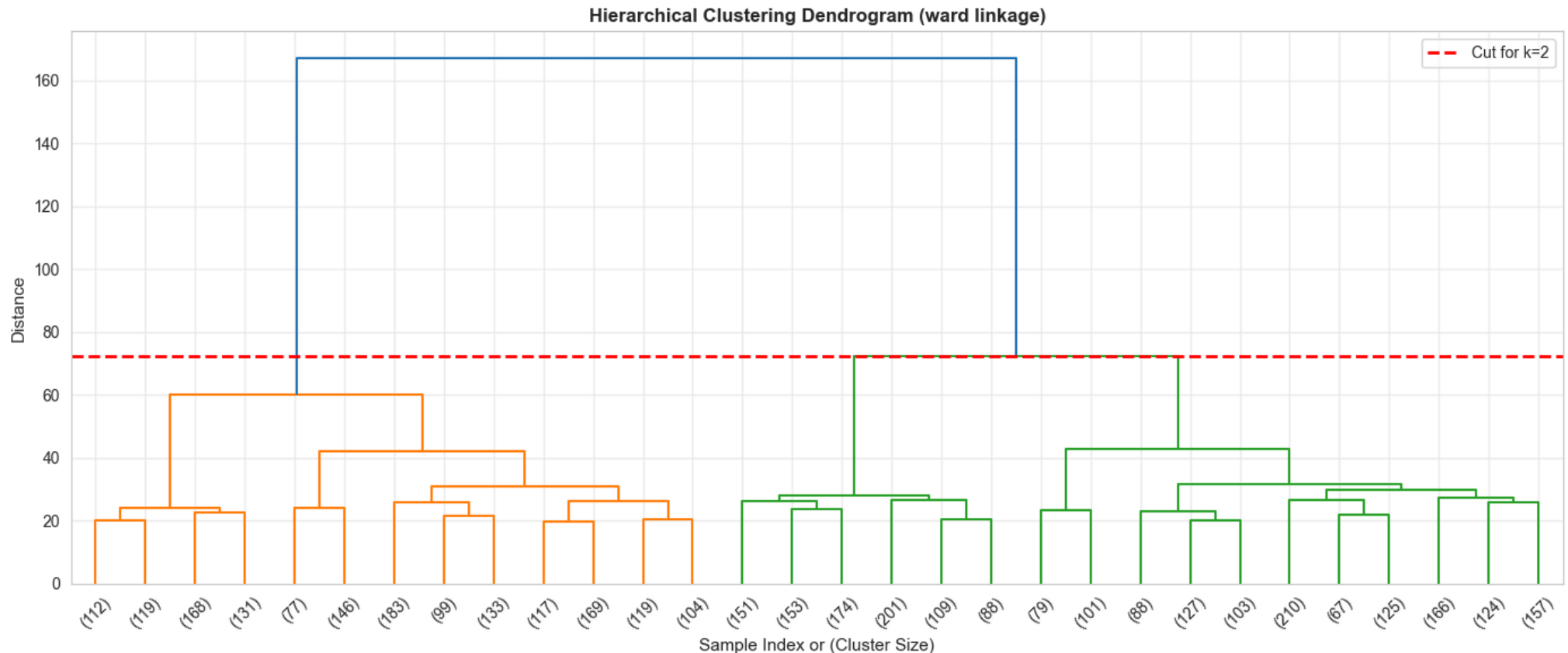
```

plt.figure(figsize=(14, 6))
dendrogram(Z, leaf_font_size=10, truncate_mode='lastp', p=30)
plt.title(f'Hierarchical Clustering Dendrogram ({best_linkage} linkage)', fontsize=12, fontweight='bold')
plt.xlabel('Sample Index or (Cluster Size)', fontsize=11)
plt.ylabel('Distance', fontsize=11)
plt.axhline(y=Z[-optimal_k, 2], c='red', linestyle='--', linewidth=2, label=f'Cut for k={optimal_k}')
plt.legend()
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()

```

Hierarchical (ward) – Silhouette Score: 0.1987
 Hierarchical (complete) – Silhouette Score: 0.1977
 Hierarchical (average) – Silhouette Score: 0.1985
 Hierarchical (single) – Silhouette Score: 0.1188

Best linkage method: ward



```

In [9]: # Apply Hierarchical Clustering with best linkage
hc_final = AgglomerativeClustering(n_clusters=optimal_k, linkage=best_linkage)

```

```

hc_clusters = hc_final.fit_predict(X_cluster)
hc_silhouette = silhouette_score(X_cluster, hc_clusters)

print(f"\nHierarchical Clustering Results (k={optimal_k}, linkage={best_linkage}):")
print(f"Silhouette Score: {hc_silhouette:.4f}")
print(f"Cluster distribution:\n{pd.Series(hc_clusters).value_counts().sort_index()}")

# Visualization
plt.figure(figsize=(12, 5))

plt.subplot(1, 2, 1)
scatter = plt.scatter(X_pca[:, 0], X_pca[:, 1], c=hc_clusters, cmap='plasma', s=50, alpha=0.6)
plt.xlabel(f'PC1 ({pca.explained_variance_ratio_[0]:.2%})', fontsize=11)
plt.ylabel(f'PC2 ({pca.explained_variance_ratio_[1]:.2%})', fontsize=11)
plt.title('Hierarchical Clustering (PCA Visualization)', fontsize=12, fontweight='bold')
plt.colorbar(scatter, label='Cluster')
plt.grid(True, alpha=0.3)

# Silhouette plot for Hierarchical
plt.subplot(1, 2, 2)
silhouette_vals_hc = silhouette_samples(X_cluster, hc_clusters)
y_lower = 10
colors = plt.cm.plasma(np.linspace(0, 1, optimal_k))

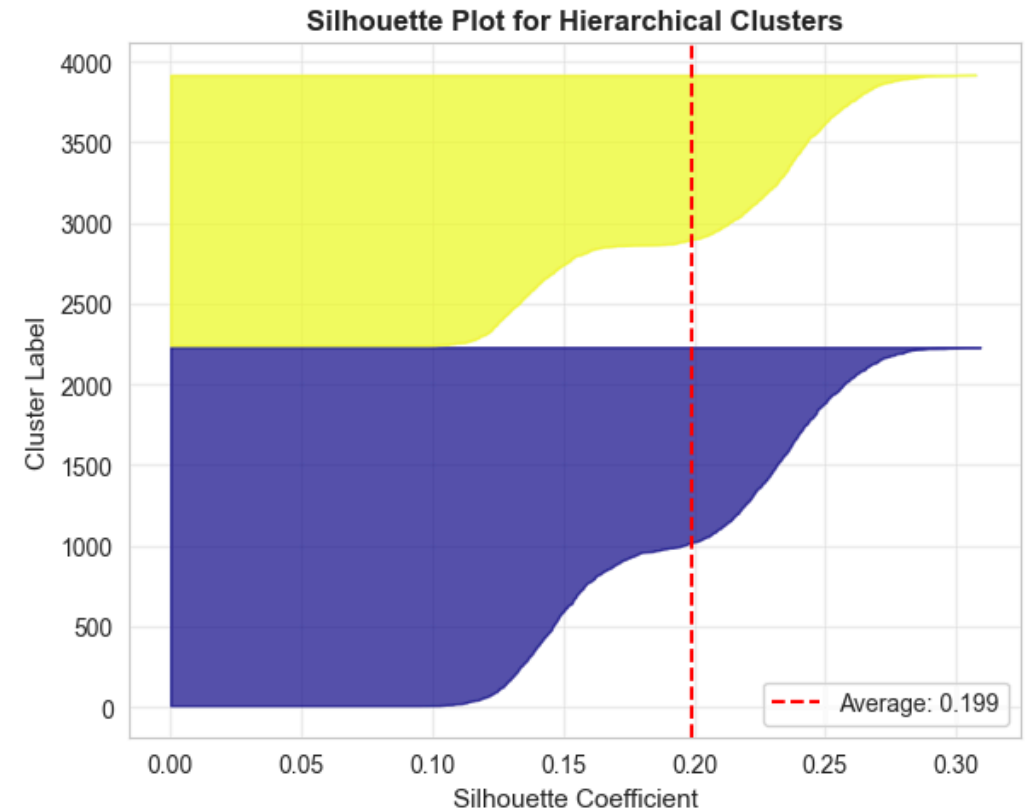
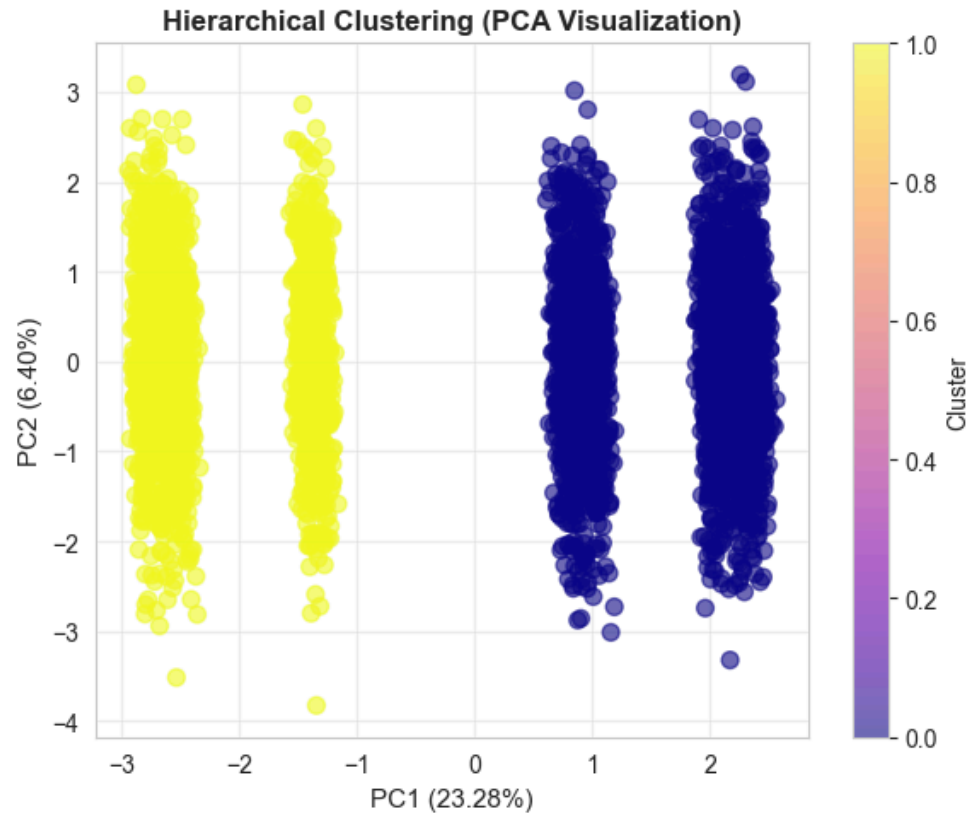
for i in range(optimal_k):
    cluster_silhouette_vals = silhouette_vals_hc[hc_clusters == i]
    cluster_silhouette_vals.sort()
    size_cluster_i = cluster_silhouette_vals.shape[0]
    y_upper = y_lower + size_cluster_i
    plt.fill_betweenx(np.arange(y_lower, y_upper), 0, cluster_silhouette_vals,
                     facecolor=colors[i], edgecolor=colors[i], alpha=0.7)
    y_lower = y_upper + 10

plt.xlabel('Silhouette Coefficient', fontsize=11)
plt.ylabel('Cluster Label', fontsize=11)
plt.title('Silhouette Plot for Hierarchical Clusters', fontsize=12, fontweight='bold')
plt.axvline(x=hc_silhouette, color='red', linestyle='--', label=f'Average: {hc_silhouette:.3f}')
plt.legend()
plt.grid(True, alpha=0.3)

plt.tight_layout()
plt.show()

```

Hierarchical Clustering Results (k=2, linkage=ward):
 Silhouette Score: 0.1987
 Cluster distribution:
 0 2223
 1 1677
 Name: count, dtype: int64



2.3 DBSCAN Clustering (1 mark)

```
In [10]: # DBSCAN parameter tuning using k-distance graph
from sklearn.neighbors import NearestNeighbors

# Calculate k-distance for k=5
neighbors = NearestNeighbors(n_neighbors=5)
neighbors_fit = neighbors.fit(X_cluster)
distances, indices = neighbors_fit.kneighbors(X_cluster)
distances = np.sort(distances[:, 4], axis=0)

plt.figure(figsize=(10, 5))
plt.plot(distances)
```



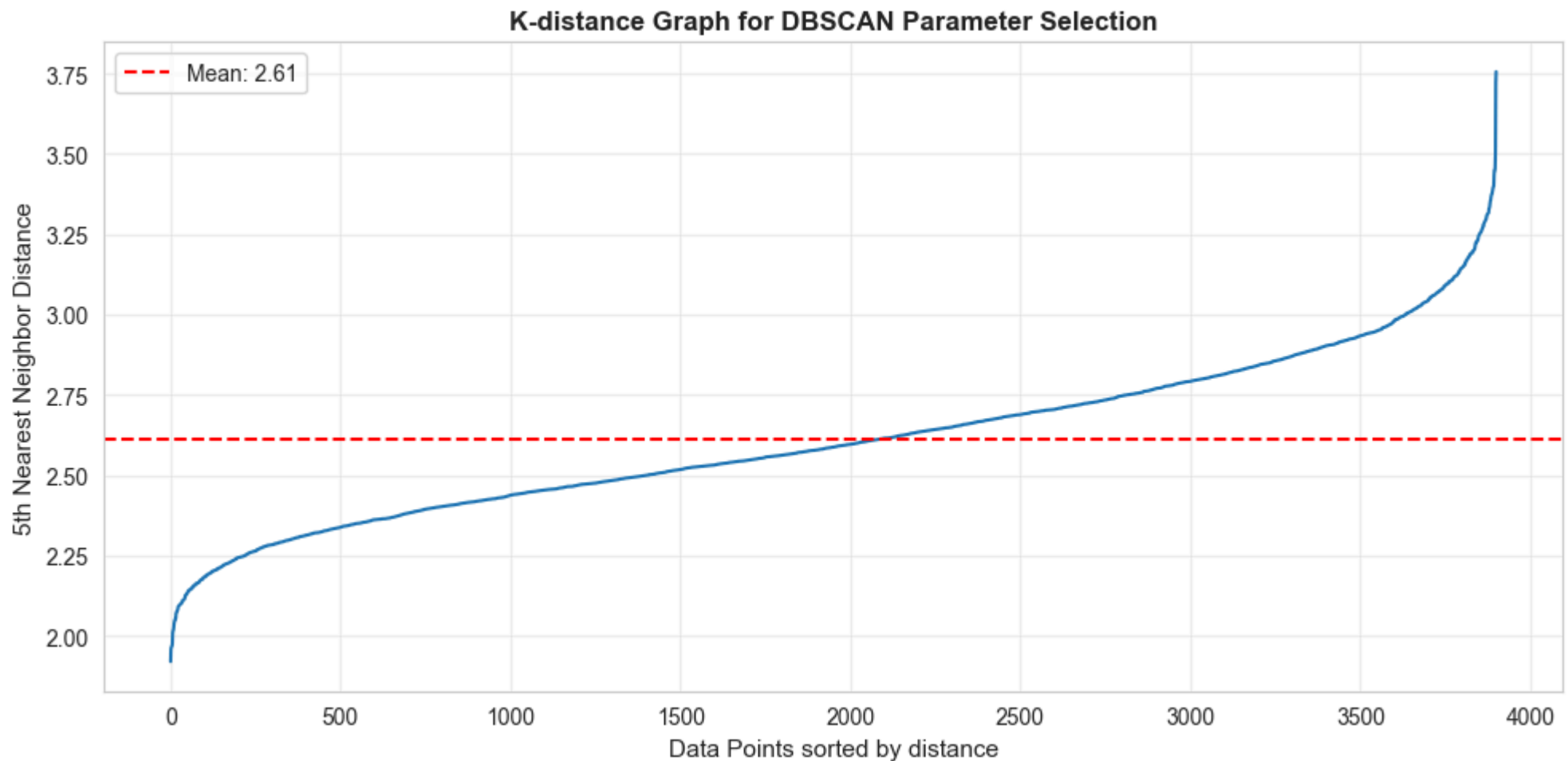
```

plt.xlabel('Data Points sorted by distance', fontsize=11)
plt.ylabel('5th Nearest Neighbor Distance', fontsize=11)
plt.title('K-distance Graph for DBSCAN Parameter Selection', fontsize=12, fontweight='bold')
plt.grid(True, alpha=0.3)
plt.axhline(y=np.mean(distances), color='r', linestyle='--', label=f'Mean: {np.mean(distances):.2f}')
plt.legend()
plt.tight_layout()
plt.show()

# Choose eps based on the elbow in the graph
eps_value = np.mean(distances)
min_samples = 5

print(f"DBSCAN Parameters: eps={eps_value:.2f}, min_samples={min_samples}")

```



DBSCAN Parameters: eps=2.61, min_samples=5

```

In [11]: # Apply DBSCAN
dbscan = DBSCAN(eps=eps_value, min_samples=min_samples)
dbscan_clusters = dbscan.fit_predict(X_cluster)

# Count clusters and noise points
n_clusters_dbscan = len(set(dbscan_clusters)) - (1 if -1 in dbscan_clusters else 0)
n_noise = list(dbscan_clusters).count(-1)

print(f"\nDBSCAN Clustering Results:")
print(f"Number of clusters: {n_clusters_dbscan}")
print(f"Number of noise points (outliers): {n_noise}")
print(f"Cluster distribution:\n{pd.Series(dbscan_clusters).value_counts().sort_index()}")

# Calculate Silhouette score (excluding noise points)
if n_clusters_dbscan > 1 and n_noise < len(dbscan_clusters):
    mask = dbscan_clusters != -1
    dbscan_silhouette = silhouette_score(X_cluster[mask], dbscan_clusters[mask])
    print(f"Silhouette Score (excluding noise): {dbscan_silhouette:.4f}")
else:
    dbscan_silhouette = -1
    print("Silhouette Score not calculable (need at least 2 clusters without all noise)")

# Visualization
plt.figure(figsize=(12, 5))

plt.subplot(1, 2, 1)
# Highlight noise points differently
colors = plt.cm.viridis(np.linspace(0, 1, n_clusters_dbscan))
for i in range(n_clusters_dbscan):
    mask = dbscan_clusters == i
    plt.scatter(X_pca[mask, 0], X_pca[mask, 1], c=[colors[i]], s=50, alpha=0.6, label=f'Cluster {i}')

# Noise points
noise_mask = dbscan_clusters == -1
plt.scatter(X_pca[noise_mask, 0], X_pca[noise_mask, 1], c='red', s=100, marker='X',
            alpha=0.8, edgecolors='black', linewidth=1.5, label='Noise')

plt.xlabel(f'PC1 ({pca.explained_variance_ratio_[0]:.2%})', fontsize=11)
plt.ylabel(f'PC2 ({pca.explained_variance_ratio_[1]:.2%})', fontsize=11)
plt.title('DBSCAN Clustering (PCA Visualization)', fontsize=12, fontweight='bold')
plt.legend(fontsize=9)
plt.grid(True, alpha=0.3)

# Cluster sizes
plt.subplot(1, 2, 2)
cluster_counts = pd.Series(dbscan_clusters).value_counts().sort_index()

```

```

colors_bar = ['red' if x == -1 else 'steelblue' for x in cluster_counts.index]
bars = plt.bar(range(len(cluster_counts)), cluster_counts.values, color=colors_bar, alpha=0.7, edgecolor='black')
plt.xlabel('Cluster Label (-1 = Noise)', fontsize=11)
plt.ylabel('Number of Points', fontsize=11)
plt.title('Cluster Size Distribution (DBSCAN)', fontsize=12, fontweight='bold')
plt.xticks(range(len(cluster_counts)), cluster_counts.index)
for i, (bar, count) in enumerate(zip(bars, cluster_counts.values)):
    plt.text(bar.get_x() + bar.get_width()/2, bar.get_height() + 5, str(count),
             ha='center', va='bottom', fontsize=10)
plt.grid(True, alpha=0.3, axis='y')

plt.tight_layout()
plt.show()

```

DBSCAN Clustering Results:

Number of clusters: 6

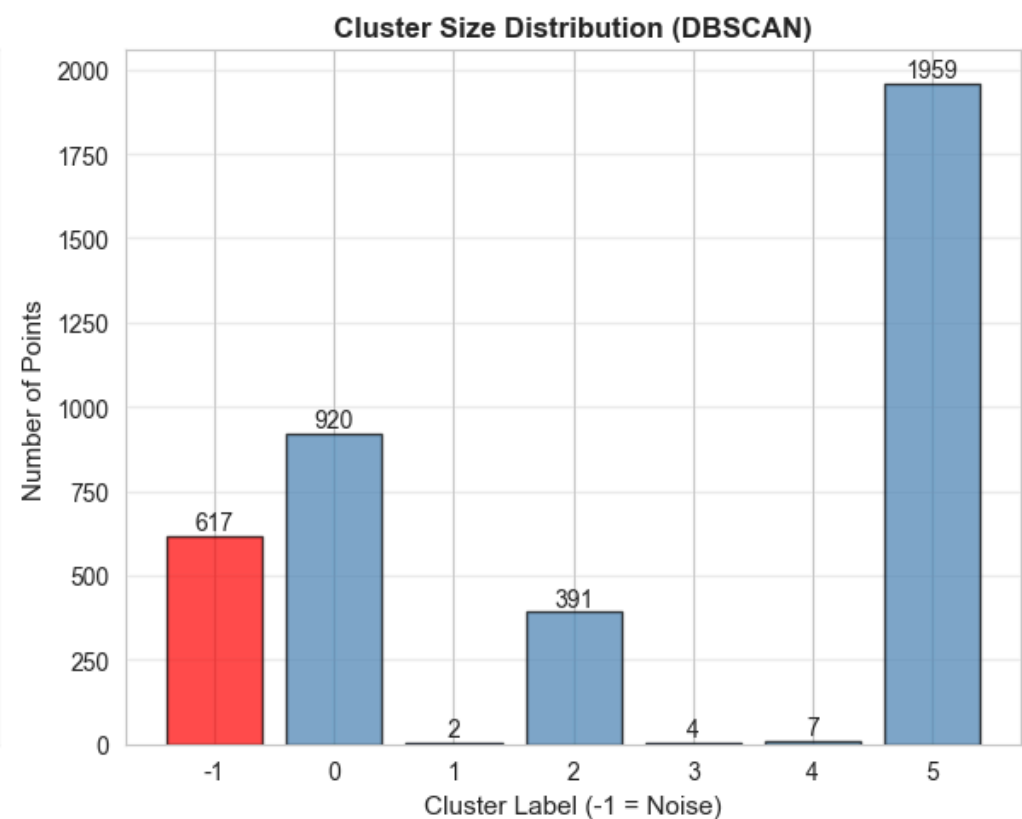
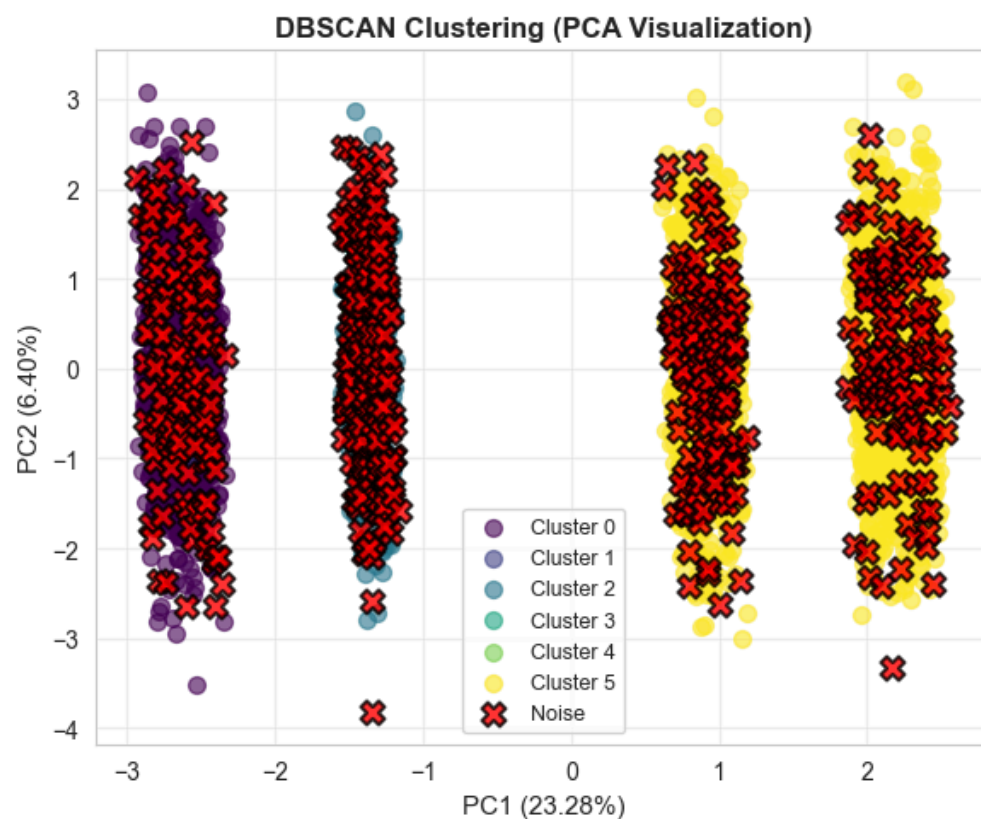
Number of noise points (outliers): 617

Cluster distribution:

-1	617
0	920
1	2
2	391
3	4
4	7
5	1959

Name: count, dtype: int64

Silhouette Score (excluding noise): 0.0612



2.4 Clustering Performance Comparison

```
In [12]: # Compare all three algorithms
comparison_data = {
    'Algorithm': ['K-Means', 'Hierarchical ('+best_linkage+')', 'DBSCAN'],
    'Silhouette Score': [kmeans_silhouette, hc_silhouette, dbscan_silhouette if dbscan_silhouette != -1 else 'N/A'],
    'Number of Clusters': [optimal_k, optimal_k, n_clusters_dbscan],
    'Noise Points': [0, 0, n_noise]
}

comparison_df = pd.DataFrame(comparison_data)
print("\n" + "="*70)
print("CLUSTERING ALGORITHMS COMPARISON")
print("="*70)
print(comparison_df.to_string(index=False))
print("="*70)

# Visualization of comparison
fig, axes = plt.subplots(1, 2, figsize=(14, 5))
```

```

# Silhouette Scores
algorithms = ['K-Means', 'Hierarchical', 'DBSCAN']
scores = [kmeans_silhouette, hc_silhouette, dbscan_silhouette if dbscan_silhouette != -1 else 0]
colors_algo = ['#2E86AB', '#A23B72', '#F18F01']

bars = axes[0].bar(algorithms, scores, color=colors_algo, alpha=0.7, edgecolor='black', linewidth=2)
axes[0].set_ylabel('Silhouette Score', fontsize=11, fontweight='bold')
axes[0].set_title('Silhouette Score Comparison', fontsize=12, fontweight='bold')
axes[0].set_ylim([0, 1])
axes[0].grid(True, alpha=0.3, axis='y')

for bar, score in zip(bars, scores):
    height = bar.get_height()
    if score != 0:
        axes[0].text(bar.get_x() + bar.get_width()/2., height + 0.02,
                     f'{score:.4f}', ha='center', va='bottom', fontsize=10, fontweight='bold')

# Number of Clusters
clusters_count = [optimal_k, optimal_k, n_clusters_dbscan]
bars2 = axes[1].bar(algorithms, clusters_count, color=colors_algo, alpha=0.7, edgecolor='black', linewidth=2)
axes[1].set_ylabel('Number of Clusters', fontsize=11, fontweight='bold')
axes[1].set_title('Cluster Count Comparison', fontsize=12, fontweight='bold')
axes[1].grid(True, alpha=0.3, axis='y')

for bar, count in zip(bars2, clusters_count):
    height = bar.get_height()
    axes[1].text(bar.get_x() + bar.get_width()/2., height + 0.1,
                 f'{count}', ha='center', va='bottom', fontsize=10, fontweight='bold')

plt.tight_layout()
plt.show()

# Determine best algorithm
print(f"\nBest performing algorithm: K-Means with Silhouette Score = {kmeans_silhouette:.4f}")

```

```

=====
CLUSTERING ALGORITHMS COMPARISON
=====

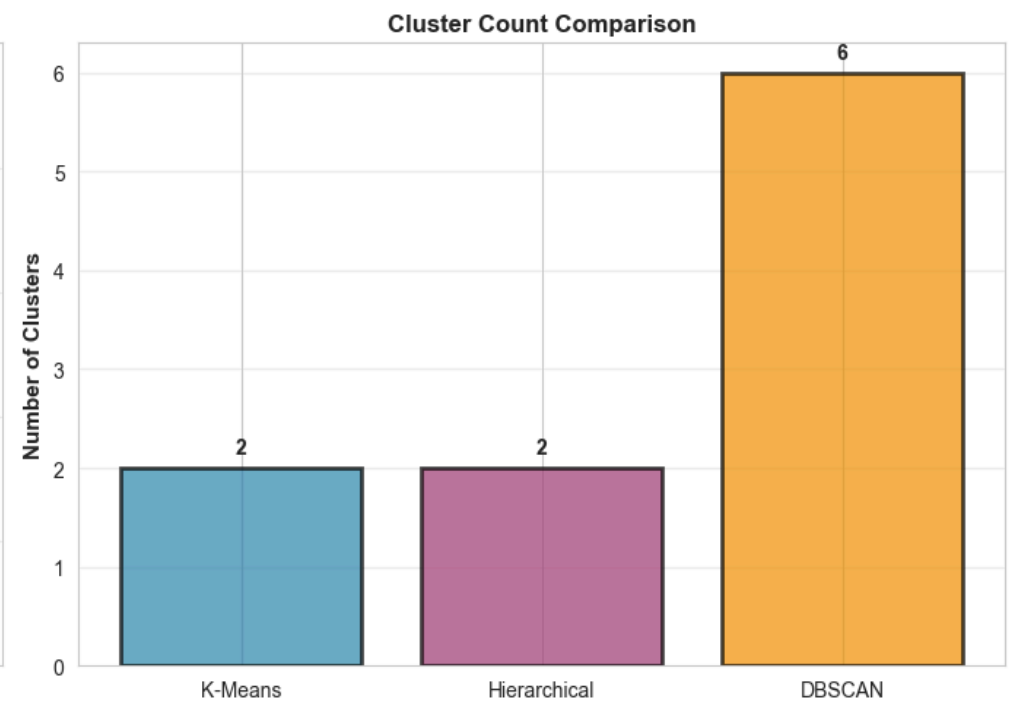
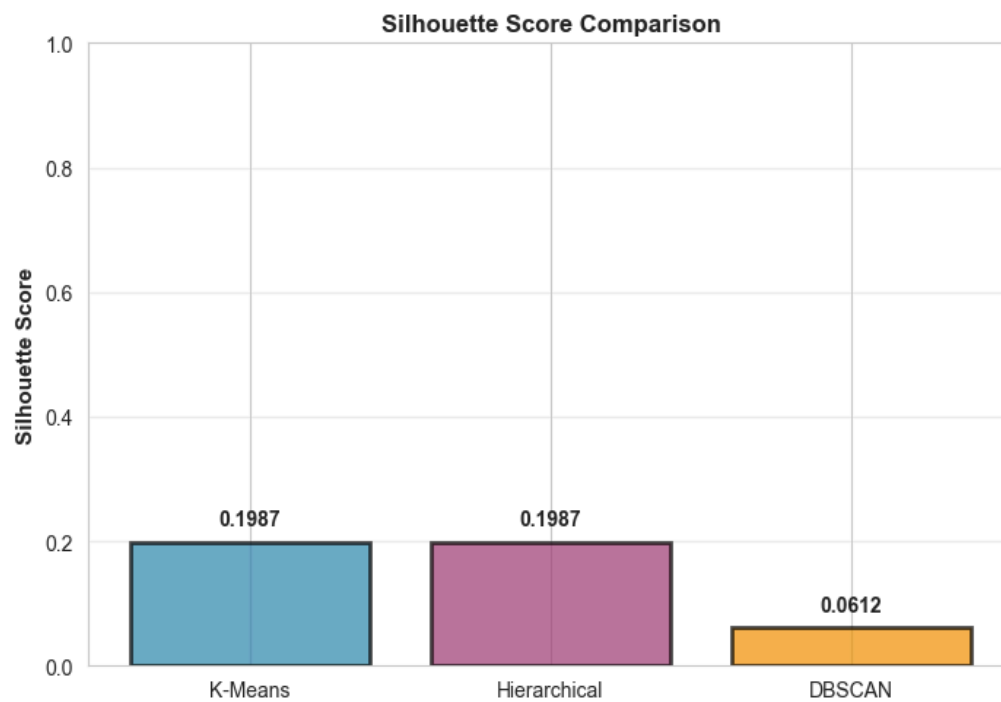
```

	Algorithm	Silhouette Score	Number of Clusters	Noise Points
	K-Means	0.198681	2	0
	Hierarchical (ward)	0.198681	2	0
	DBSCAN	0.061207	6	617

```

=====

```



Best performing algorithm: K-Means with Silhouette Score = 0.1987

3. Shopping Pattern Analysis (5 marks)

Question 3.1: Which payment method is preferred by the majority of customers?

```
In [13]: # Question 3.1: Payment Method Analysis
payment_counts = df_clean['Payment Method'].value_counts()
payment_percentage = (payment_counts / len(df_clean) * 100).round(2)

print("\n" + "="*70)
print("QUESTION 3.1: Payment Method Preference")
print("="*70)
print(f"\nPayment Method Distribution:")
print(payment_counts)
print(f"\nPercentage Distribution:")
print(payment_percentage)

majority_payment = payment_counts.index[0]
majority_count = payment_counts.values[0]
majority_percentage = payment_percentage.values[0]
```

```

print(f"\nAnswer: The most preferred payment method is '{majority_payment}'")
print(f"Number of customers: {majority_count}")
print(f"Percentage of total customers: {majority_percentage}%")

# Visualization
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# Bar chart
colors_payment = plt.cm.Set3(np.linspace(0, 1, len(payment_counts)))
bars = axes[0].bar(range(len(payment_counts)), payment_counts.values, color=colors_payment,
                  alpha=0.7, edgecolor='black', linewidth=1.5)
axes[0].set_xticks(range(len(payment_counts)))
axes[0].set_xticklabels(payment_counts.index, rotation=45, ha='right')
axes[0].set_ylabel('Number of Customers', fontsize=11, fontweight='bold')
axes[0].set_title('Payment Method Distribution', fontsize=12, fontweight='bold')
axes[0].grid(True, alpha=0.3, axis='y')

for bar in bars:
    height = bar.get_height()
    axes[0].text(bar.get_x() + bar.get_width()/2., height,
                f'{int(height)}', ha='center', va='bottom', fontsize=9)

# Pie chart
axes[1].pie(payment_counts.values, labels=payment_counts.index, autopct='%1.1f%%',
            colors=colors_payment, startangle=90, textprops={'fontsize': 10})
axes[1].set_title('Payment Method Percentage Distribution', fontsize=12, fontweight='bold')

plt.tight_layout()
plt.show()

```

```
=====
QUESTION 3.1: Payment Method Preference
=====
```

Payment Method Distribution:

Payment Method

PayPal 677

Credit Card 671

Cash 670

Debit Card 636

Venmo 634

Bank Transfer 612

Name: count, dtype: int64

Percentage Distribution:

Payment Method

PayPal 17.36

Credit Card 17.21

Cash 17.18

Debit Card 16.31

Venmo 16.26

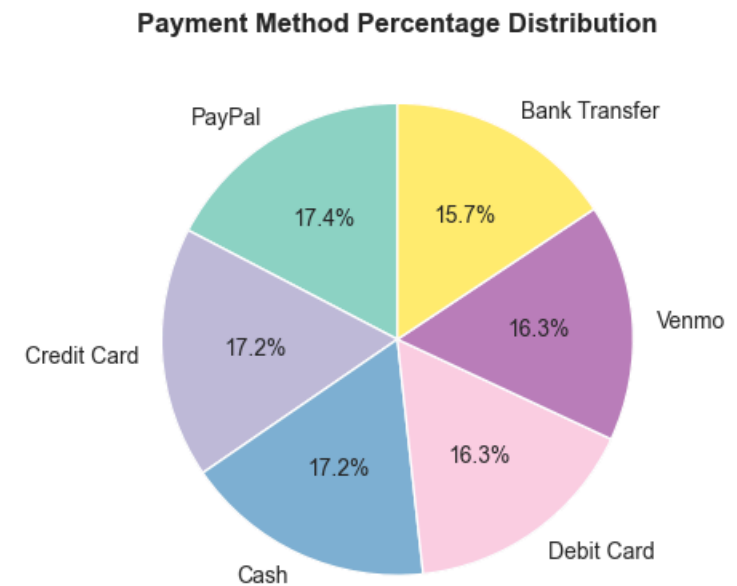
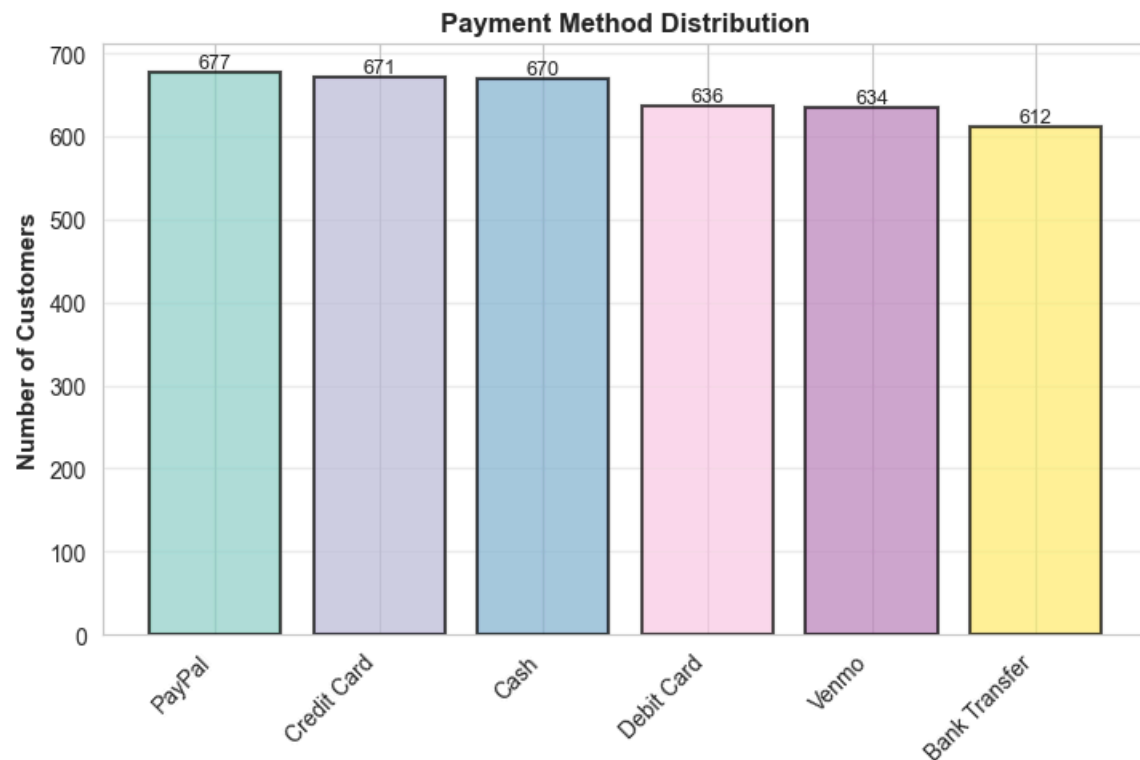
Bank Transfer 15.69

Name: count, dtype: float64

Answer: The most preferred payment method is 'PayPal'

Number of customers: 677

Percentage of total customers: 17.36%



Question 3.2: What category is most frequently purchased based on shopping behavior, and how does it vary by gender?

```
In [14]: # Question 3.2: Category Purchase Analysis by Gender
category_counts = df_clean['Category'].value_counts()
most_frequent_category = category_counts.index[0]
most_frequent_count = category_counts.values[0]

print("\n" + "="*70)
print("QUESTION 3.2: Category Purchase Frequency and Gender Variation")
print("="*70)
print(f"\nOverall Category Distribution:")
print(category_counts)

print(f"\nMost frequently purchased category: {most_frequent_category}")
print(f"Purchase count: {most_frequent_count}")

# Category by Gender
category_by_gender = pd.crosstab(df_clean['Category'], df_clean['Gender'])
category_by_gender_pct = category_by_gender.div(category_by_gender.sum(axis=0), axis=1) * 100
```

```

print(f"\nCategory Distribution by Gender (Count):")
print(category_by_gender)
print(f"\nCategory Distribution by Gender (Percentage):")
print(category_by_gender_pct.round(2))

# Visualization
fig, axes = plt.subplots(2, 2, figsize=(14, 10))

# Overall category distribution
colors_cat = plt.cm.Spectral(np.linspace(0, 1, len(category_counts)))
axes[0, 0].bar(range(len(category_counts)), category_counts.values, color=colors_cat,
               alpha=0.7, edgecolor='black', linewidth=1.5)
axes[0, 0].set_xticks(range(len(category_counts)))
axes[0, 0].set_xticklabels(category_counts.index, rotation=45, ha='right')
axes[0, 0].set_ylabel('Number of Purchases', fontsize=11, fontweight='bold')
axes[0, 0].set_title('Overall Category Purchase Frequency', fontsize=12, fontweight='bold')
axes[0, 0].grid(True, alpha=0.3, axis='y')

# Grouped bar chart by gender
x = np.arange(len(category_counts))
width = 0.35
for i, gender in enumerate(category_by_gender.columns):
    axes[0, 1].bar(x + i*width, category_by_gender[gender].values, width,
                  label=gender, alpha=0.7, edgecolor='black', linewidth=1.5)

axes[0, 1].set_xlabel('Category', fontsize=11, fontweight='bold')
axes[0, 1].set_ylabel('Number of Purchases', fontsize=11, fontweight='bold')
axes[0, 1].set_title('Category Purchases by Gender', fontsize=12, fontweight='bold')
axes[0, 1].set_xticks(x + width/2)
axes[0, 1].set_xticklabels(category_counts.index, rotation=45, ha='right')
axes[0, 1].legend()
axes[0, 1].grid(True, alpha=0.3, axis='y')

# Stacked bar chart percentage
category_by_gender_pct.T.plot(kind='bar', stacked=True, ax=axes[1, 0],
                             color=colors_cat, alpha=0.7, edgecolor='black', linewidth=1.5)
axes[1, 0].set_ylabel('Percentage', fontsize=11, fontweight='bold')
axes[1, 0].set_title('Category Percentage by Gender (Stacked)', fontsize=12, fontweight='bold')
axes[1, 0].set_xticklabels(axes[1, 0].get_xticklabels(), rotation=0)
axes[1, 0].legend(title='Category', bbox_to_anchor=(1.05, 1), loc='upper left', fontsize=9)
axes[1, 0].grid(True, alpha=0.3, axis='y')

# Heatmap
sns.heatmap(category_by_gender_pct, annot=True, fmt='.1f', cmap='YlOrRd',
            ax=axes[1, 1], cbar_kws={'label': 'Percentage (%)'}, linewidths=1, linecolor='black')

```

```

axes[1, 1].set_title('Category Purchase % Heatmap by Gender', fontsize=12, fontweight='bold')
axes[1, 1].set_ylabel('Category', fontsize=11, fontweight='bold')
axes[1, 1].set_xlabel('Gender', fontsize=11, fontweight='bold')

plt.tight_layout()
plt.show()

print(f"\nAnswer: The most frequently purchased category is '{most_frequent_category}'.")
print(f"Across genders, {most_frequent_category} shows:")
for gender in category_by_gender.columns:
    pct = category_by_gender_pct.loc[most_frequent_category, gender]
    count = category_by_gender.loc[most_frequent_category, gender]
    print(f" - {gender}: {count} purchases ({pct:.1f}% of {gender} purchases)")

```

QUESTION 3.2: Category Purchase Frequency and Gender Variation

Overall Category Distribution:

Category	
Clothing	1737
Accessories	1240
Footwear	599
Outerwear	324

Name: count, dtype: int64

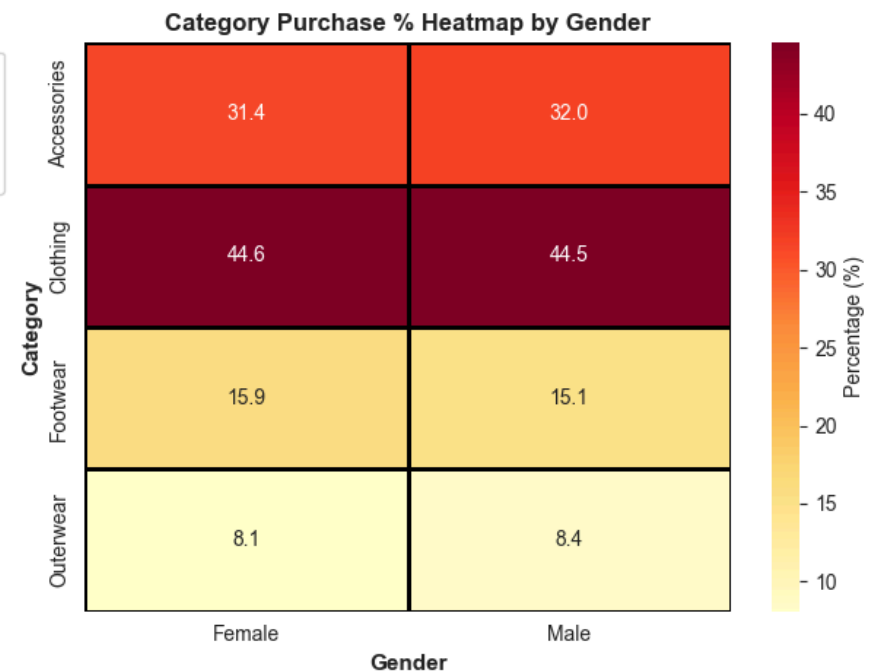
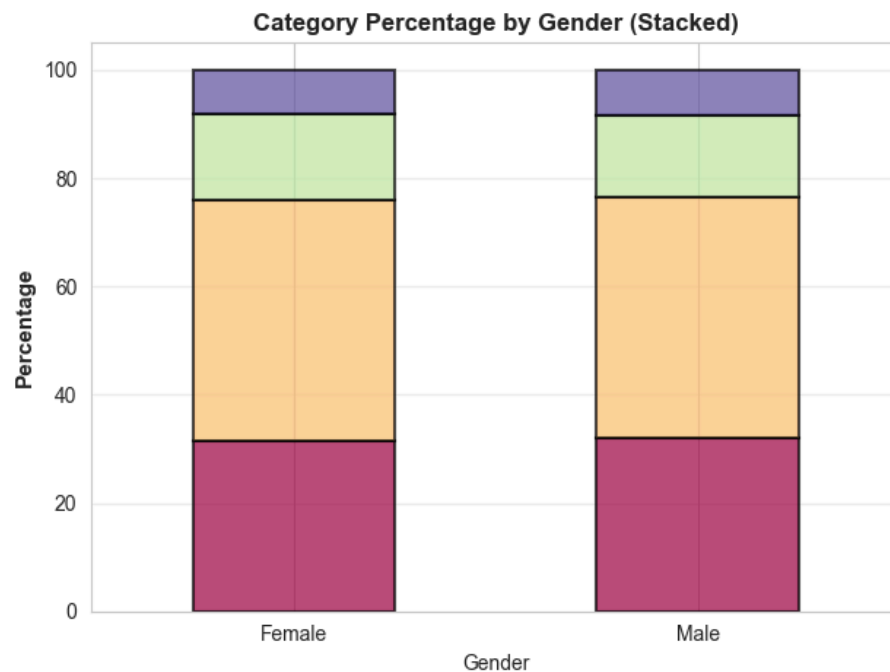
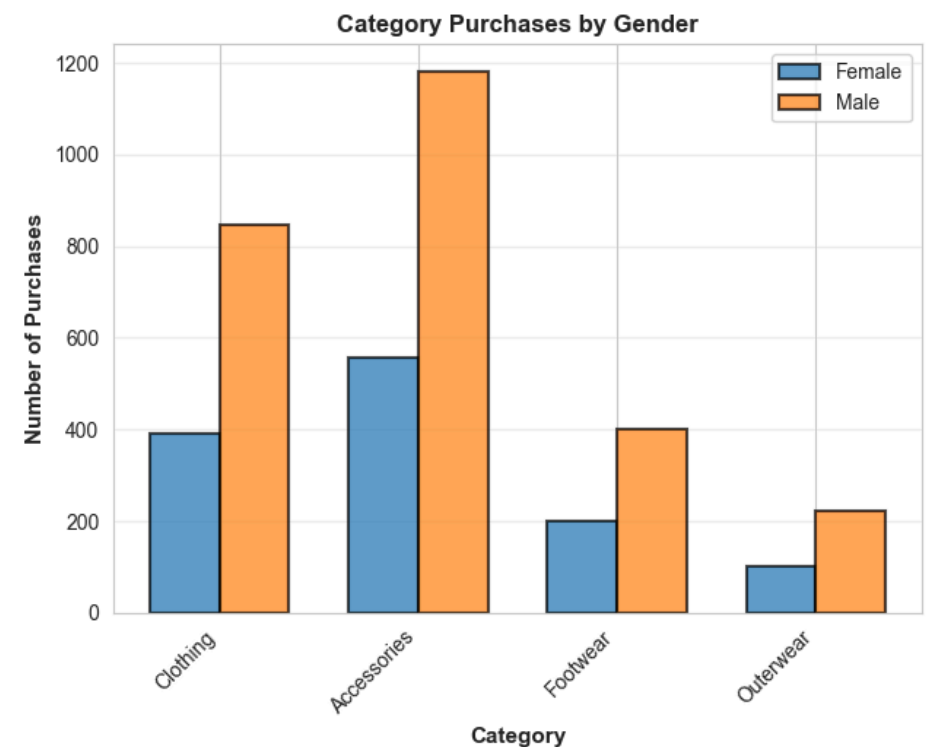
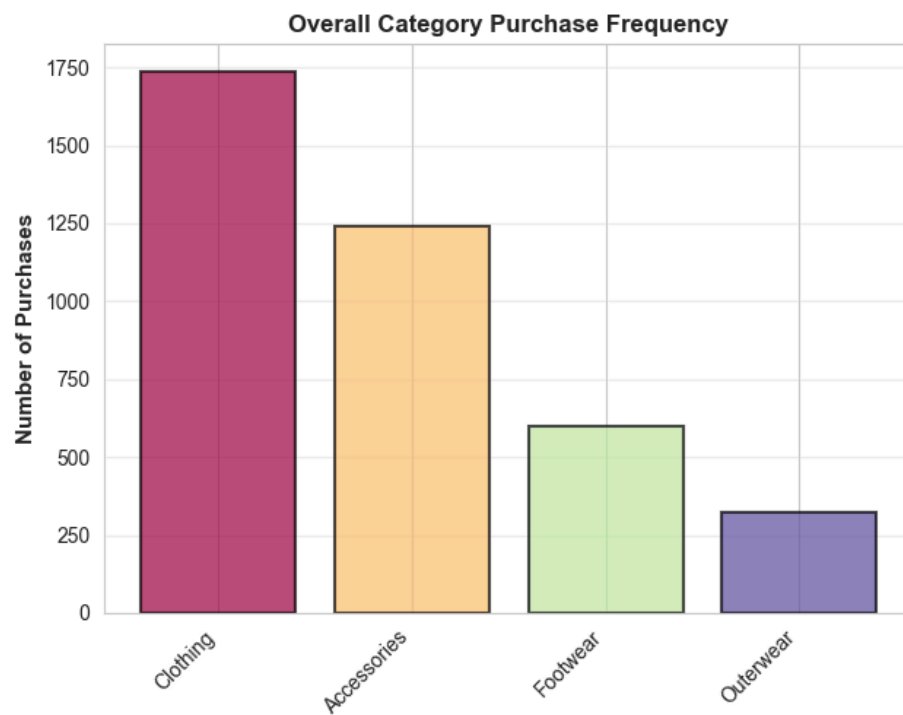
Most frequently purchased category: Clothing
Purchase count: 1737

Category Distribution by Gender (Count):

Gender	Female	Male
Category		
Accessories	392	848
Clothing	556	1181
Footwear	199	400
Outerwear	101	223

Category Distribution by Gender (Percentage):

Gender	Female	Male
Category		
Accessories	31.41	31.98
Clothing	44.55	44.53
Footwear	15.95	15.08
Outerwear	8.09	8.41



Answer: The most frequently purchased category is 'Clothing'.

Across genders, Clothing shows:

- Female: 556 purchases (44.6% of Female purchases)
- Male: 1181 purchases (44.5% of Male purchases)

Question 3.3: Is there a notable difference in review ratings between male and female customers when making purchases?

```
In [15]: # Question 3.3: Review Ratings by Gender
from scipy import stats

print("\n" + "="*70)
print("QUESTION 3.3: Review Rating Differences by Gender")
print("="*70)

male_ratings = df_clean[df_clean['Gender'] == 'Male']['Review Rating']
female_ratings = df_clean[df_clean['Gender'] == 'Female']['Review Rating']

print(f"\nMale Customers - Review Ratings:")
print(f"  Count: {len(male_ratings)}")
print(f"  Mean: {male_ratings.mean():.4f}")
print(f"  Median: {male_ratings.median():.4f}")
print(f"  Std Dev: {male_ratings.std():.4f}")
print(f"  Min: {male_ratings.min()}, Max: {male_ratings.max()}")

print(f"\nFemale Customers - Review Ratings:")
print(f"  Count: {len(female_ratings)}")
print(f"  Mean: {female_ratings.mean():.4f}")
print(f"  Median: {female_ratings.median():.4f}")
print(f"  Std Dev: {female_ratings.std():.4f}")
print(f"  Min: {female_ratings.min()}, Max: {female_ratings.max()}")

# Statistical test - Independent t-test
t_stat, p_value = stats.ttest_ind(male_ratings, female_ratings)
print(f"\nIndependent t-test:")
print(f"  t-statistic: {t_stat:.4f}")
print(f"  p-value: {p_value:.4f}")

if p_value < 0.05:
    print(f"  Result: SIGNIFICANT difference (p < 0.05)")
else:
    print(f"  Result: NO significant difference (p >= 0.05)")

# Distribution comparison
print(f"\nRating Distribution by Gender:")
rating_dist = pd.crosstab(df_clean['Gender'], df_clean['Review Rating'])
print(rating_dist)
```

```

# Visualization
fig, axes = plt.subplots(2, 2, figsize=(14, 10))

# Box plot
box_data = [male_ratings, female_ratings]
bp = axes[0, 0].boxplot(box_data, labels=['Male', 'Female'], patch_artist=True)
for patch, color in zip(bp['boxes'], ['lightblue', 'lightpink']):
    patch.set_facecolor(color)
axes[0, 0].set_ylabel('Review Rating', fontsize=11, fontweight='bold')
axes[0, 0].set_title('Review Rating Distribution by Gender (Box Plot)', fontsize=12, fontweight='bold')
axes[0, 0].grid(True, alpha=0.3, axis='y')

# Violin plot
parts = axes[0, 1].violinplot([male_ratings, female_ratings], positions=[1, 2], showmeans=True, showmedians=True)
axes[0, 1].set_xticks([1, 2])
axes[0, 1].set_xticklabels(['Male', 'Female'])
axes[0, 1].set_ylabel('Review Rating', fontsize=11, fontweight='bold')
axes[0, 1].set_title('Review Rating Distribution (Violin Plot)', fontsize=12, fontweight='bold')
axes[0, 1].grid(True, alpha=0.3, axis='y')

# Histogram
axes[1, 0].hist(male_ratings, bins=20, alpha=0.6, label='Male', color='blue', edgecolor='black')
axes[1, 0].hist(female_ratings, bins=20, alpha=0.6, label='Female', color='red', edgecolor='black')
axes[1, 0].set_xlabel('Review Rating', fontsize=11, fontweight='bold')
axes[1, 0].set_ylabel('Frequency', fontsize=11, fontweight='bold')
axes[1, 0].set_title('Review Rating Histogram by Gender', fontsize=12, fontweight='bold')
axes[1, 0].legend()
axes[1, 0].grid(True, alpha=0.3, axis='y')

# Bar chart of distribution
rating_dist_pct = rating_dist.div(rating_dist.sum(axis=1), axis=0) * 100
rating_dist_pct.T.plot(kind='bar', ax=axes[1, 1], color=['lightblue', 'lightpink'],
                        alpha=0.7, edgecolor='black', linewidth=1.5)
axes[1, 1].set_ylabel('Percentage', fontsize=11, fontweight='bold')
axes[1, 1].set_title('Rating Distribution % by Gender', fontsize=12, fontweight='bold')
axes[1, 1].set_xticklabels(axes[1, 1].get_xticklabels(), rotation=0)
axes[1, 1].legend(title='Gender')
axes[1, 1].grid(True, alpha=0.3, axis='y')

plt.tight_layout()
plt.show()

mean_diff = abs(male_ratings.mean() - female_ratings.mean())
print(f"\nAnswer: Mean rating difference: {mean_diff:.4f}")
if p_value < 0.05:
    print(f"There IS a statistically significant difference in review ratings between genders (p={p_value:.4f})")

```

```
else:
    print(f"There is NO statistically significant difference in review ratings between genders (p={p_value:.4f})")
```

QUESTION 3.3: Review Rating Differences by Gender

Male Customers – Review Ratings:

Count: 2652
Mean: 3.7540
Median: 3.8000
Std Dev: 0.7142
Min: 2.5, Max: 5.0

Female Customers – Review Ratings:

Count: 1248
Mean: 3.7414
Median: 3.7000
Std Dev: 0.7206
Min: 2.5, Max: 5.0

Independent t-test:

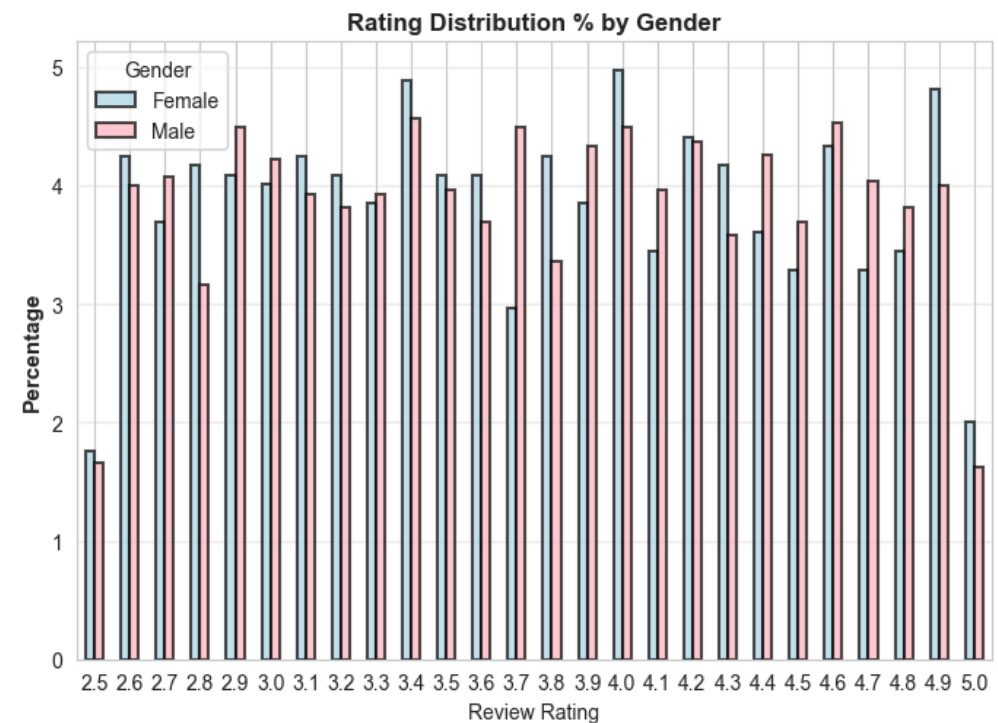
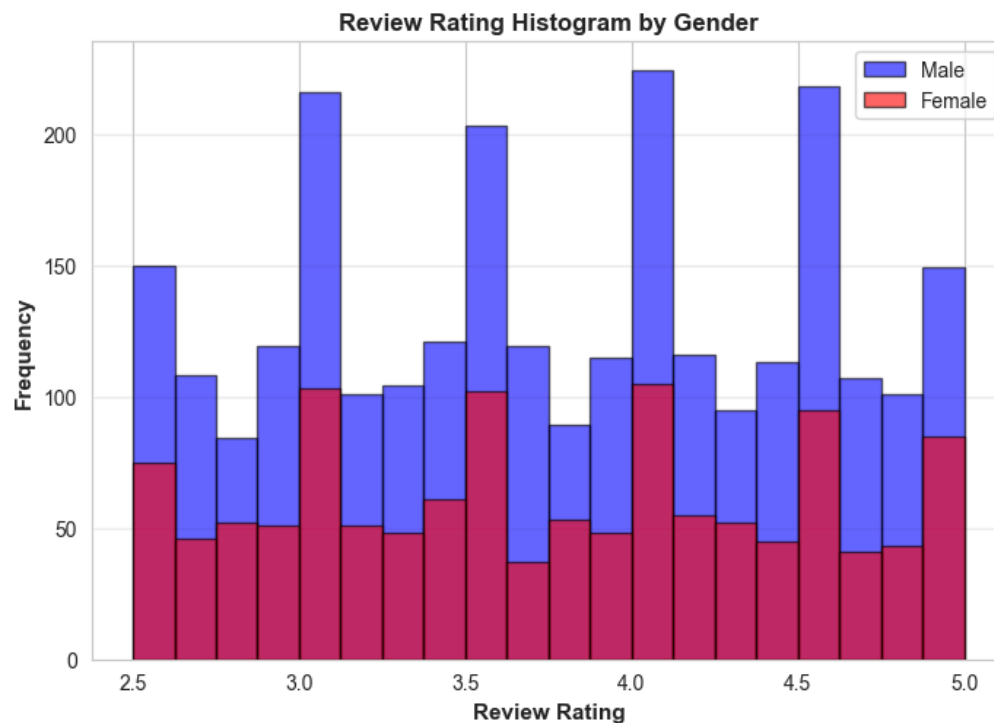
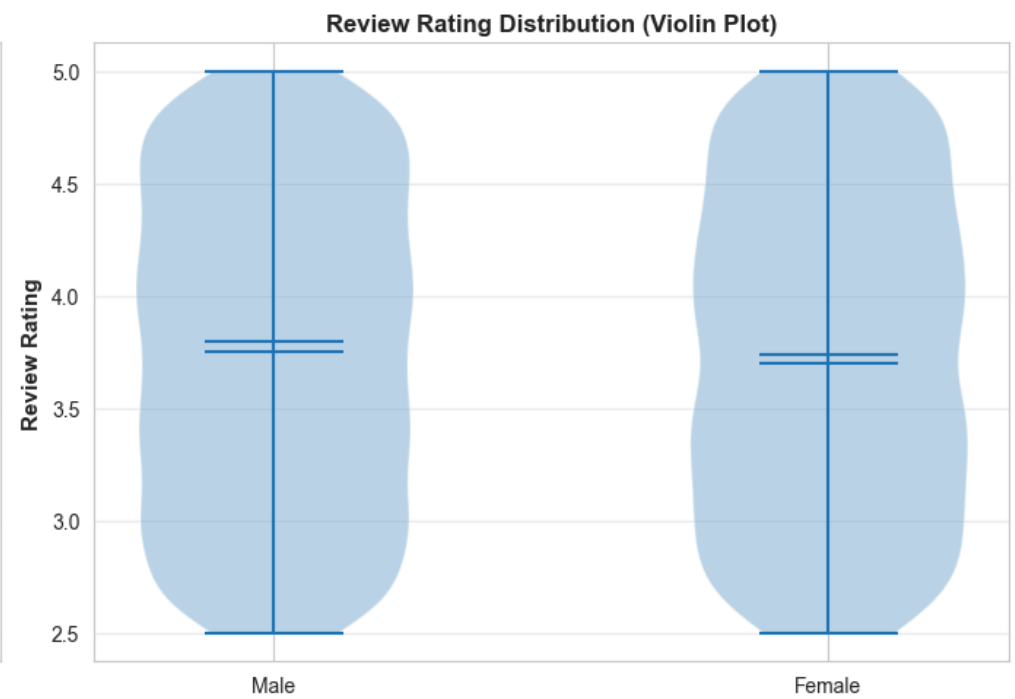
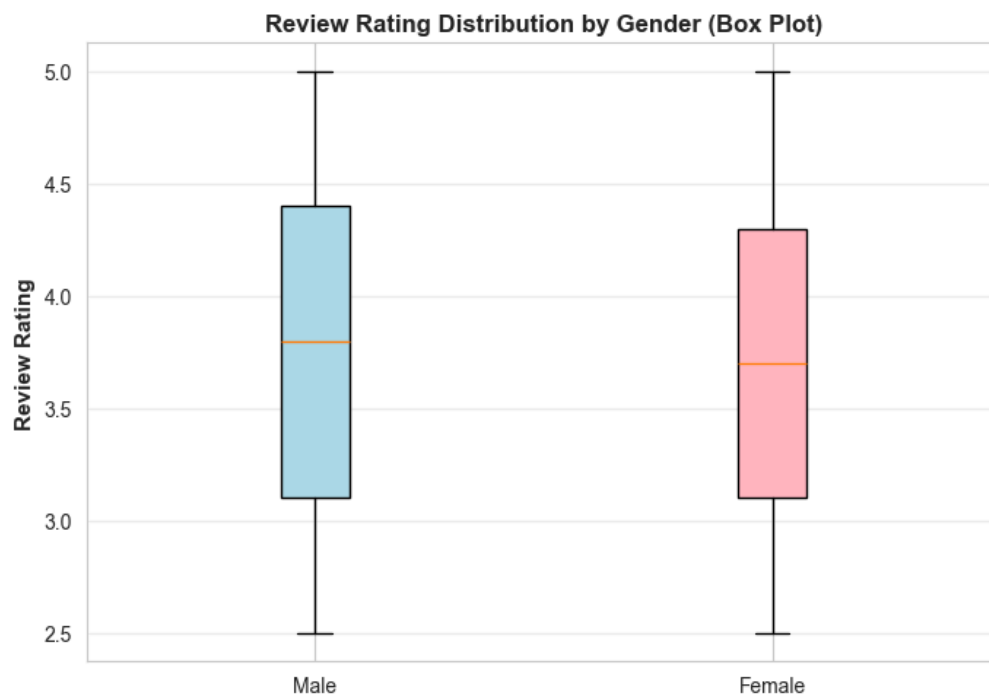
t-statistic: 0.5097
p-value: 0.6103
Result: NO significant difference ($p \geq 0.05$)

Rating Distribution by Gender:

Review Rating	2.5	2.6	2.7	2.8	2.9	3.0	3.1	3.2	3.3	3.4	...	4.1	\
Gender											...		
Female	22	53	46	52	51	50	53	51	48	61	...	43	
Male	44	106	108	84	119	112	104	101	104	121	...	105	

Review Rating	4.2	4.3	4.4	4.5	4.6	4.7	4.8	4.9	5.0
Gender									
Female	55	52	45	41	54	41	43	60	25
Male	116	95	113	98	120	107	101	106	43

[2 rows x 26 columns]



Answer: Mean rating difference: 0.0125

There is NO statistically significant difference in review ratings between genders ($p=0.6103$)

Question 3.4: What state has the lowest purchase amount, and which state tends to make higher purchases?

```
In [16]: # Question 3.4: State-wise Purchase Analysis
print("\n" + "="*70)
print("QUESTION 3.4: State-wise Purchase Amount Analysis")
print("="*70)

state_purchases = df_clean.groupby('Location')['Purchase Amount (USD)'].agg(['sum', 'mean', 'count']).round(2)
state_purchases = state_purchases.sort_values('sum', ascending=False)

print(f"\nState-wise Purchase Statistics:")
print(state_purchases.head(10))

lowest_state = state_purchases['sum'].idxmin()
lowest_amount = state_purchases['sum'].min()
highest_state = state_purchases['sum'].idxmax()
highest_amount = state_purchases['sum'].max()

print(f"\nState with LOWEST total purchases: {lowest_state}")
print(f"  Total Purchase Amount: ${lowest_amount:,.2f}")
print(f"  Average per transaction: ${state_purchases.loc[lowest_state, 'mean']:.2f}")
print(f"  Number of transactions: {int(state_purchases.loc[lowest_state, 'count'])}")

print(f"\nState with HIGHEST total purchases: {highest_state}")
print(f"  Total Purchase Amount: ${highest_amount:,.2f}")
print(f"  Average per transaction: ${state_purchases.loc[highest_state, 'mean']:.2f}")
print(f"  Number of transactions: {int(state_purchases.loc[highest_state, 'count'])}")

# Visualization
fig, axes = plt.subplots(2, 2, figsize=(16, 10))

# Top 15 states by total purchase amount
top15_states = state_purchases.head(15)
colors_state = plt.cm.RdYlGn(np.linspace(0.3, 0.7, len(top15_states)))
bars1 = axes[0, 0].barh(range(len(top15_states)), top15_states['sum'], color=colors_state,
                        alpha=0.7, edgecolor='black', linewidth=1.5)
axes[0, 0].set_yticks(range(len(top15_states)))
axes[0, 0].set_yticklabels(top15_states.index)
axes[0, 0].set_xlabel('Total Purchase Amount ($)', fontsize=11, fontweight='bold')
axes[0, 0].set_title('Top 15 States by Total Purchase Amount', fontsize=12, fontweight='bold')
axes[0, 0].grid(True, alpha=0.3, axis='x')
for i, bar in enumerate(bars1):
    width = bar.get_width()
```

```

        axes[0, 0].text(width, bar.get_y() + bar.get_height()/2.,
                        f'${width:,.0f}', ha='left', va='center', fontsize=9)

# Bottom 10 states by total purchase amount
bottom10_states = state_purchases.tail(10)
colors_bottom = plt.cm.RdYlGn(np.linspace(0.2, 0.4, len(bottom10_states)))
bars2 = axes[0, 1].barh(range(len(bottom10_states)), bottom10_states['sum'], color=colors_bottom,
                        alpha=0.7, edgecolor='black', linewidth=1.5)
axes[0, 1].set_yticks(range(len(bottom10_states)))
axes[0, 1].set_yticklabels(bottom10_states.index)
axes[0, 1].set_xlabel('Total Purchase Amount ($)', fontsize=11, fontweight='bold')
axes[0, 1].set_title('Bottom 10 States by Total Purchase Amount', fontsize=12, fontweight='bold')
axes[0, 1].grid(True, alpha=0.3, axis='x')
for i, bar in enumerate(bars2):
    width = bar.get_width()
    axes[0, 1].text(width, bar.get_y() + bar.get_height()/2.,
                    f'${width:,.0f}', ha='left', va='center', fontsize=9)

# Average purchase by state - Top 15
top15_avg = state_purchases.head(15).sort_values('mean', ascending=False)
colors_avg = plt.cm.coolwarm(np.linspace(0.2, 0.8, len(top15_avg)))
axes[1, 0].bar(range(len(top15_avg)), top15_avg['mean'], color=colors_avg, alpha=0.7, edgecolor='black', linewidth=1.5)
axes[1, 0].set_xticks(range(len(top15_avg)))
axes[1, 0].set_xticklabels(top15_avg.index, rotation=45, ha='right')
axes[1, 0].set_ylabel('Average Purchase Amount ($)', fontsize=11, fontweight='bold')
axes[1, 0].set_title('Top 15 States by Average Purchase Amount', fontsize=12, fontweight='bold')
axes[1, 0].grid(True, alpha=0.3, axis='y')

# Scatter: Average vs Total for all states
scatter = axes[1, 1].scatter(state_purchases['count'], state_purchases['sum'],
                             s=state_purchases['mean']*10, alpha=0.6, c=state_purchases['mean'],
                             cmap='viridis', edgecolors='black', linewidth=1.5)
axes[1, 1].set_xlabel('Number of Transactions', fontsize=11, fontweight='bold')
axes[1, 1].set_ylabel('Total Purchase Amount ($)', fontsize=11, fontweight='bold')
axes[1, 1].set_title('State Analysis: Transaction Count vs Total Amount', fontsize=12, fontweight='bold')
axes[1, 1].grid(True, alpha=0.3)
plt.colorbar(scatter, ax=axes[1, 1], label='Avg Purchase ($)')

# Annotate highest and lowest
axes[1, 1].annotate(highest_state, (state_purchases.loc[highest_state, 'count'],
                                     state_purchases.loc[highest_state, 'sum']),
                    xytext=(10, 10), textcoords='offset points', fontsize=9,
                    bbox=dict(boxstyle='round,pad=0.3', facecolor='yellow', alpha=0.7),
                    arrowprops=dict(arrowstyle='->', connectionstyle='arc3,rad=0'))

plt.tight_layout()

```

```
plt.show()

print(f"\nAnswer: State with lowest purchase amount: {lowest_state} (${lowest_amount:,.2f})")
print(f"Answer: State with highest purchase amount: {highest_state} (${highest_amount:,.2f})")
print(f"Difference: ${highest_amount - lowest_amount:,.2f}")
```

QUESTION 3.4: State-wise Purchase Amount Analysis

State-wise Purchase Statistics:

	sum	mean	count
Location			
Montana	5784	60.25	96
Illinois	5617	61.05	92
California	5605	59.00	95
Idaho	5587	60.08	93
Nevada	5514	63.38	87
Alabama	5261	59.11	89
New York	5257	60.43	87
North Dakota	5220	62.89	83
West Virginia	5174	63.88	81
Nebraska	5172	59.45	87

State with LOWEST total purchases: Kansas

Total Purchase Amount: \$3,437.00

Average per transaction: \$54.56

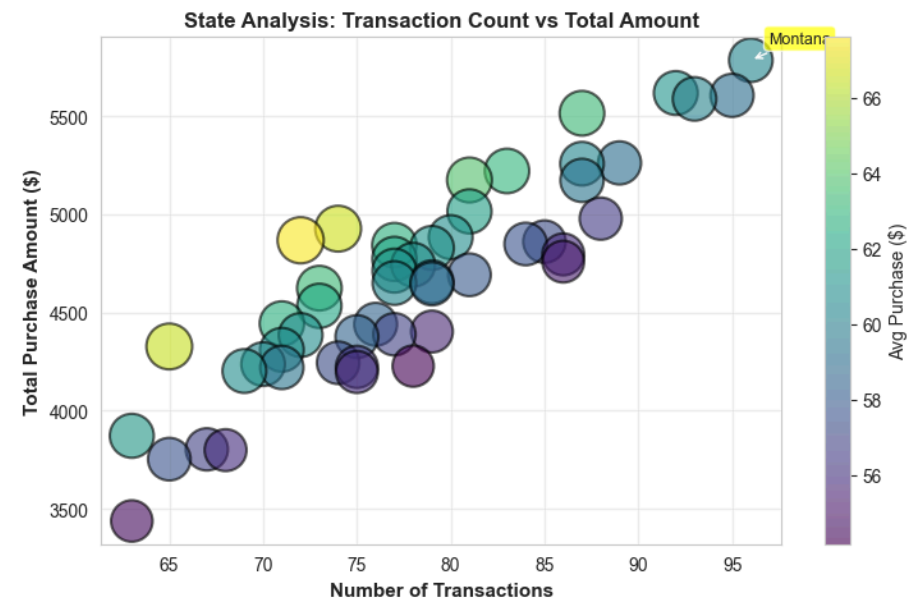
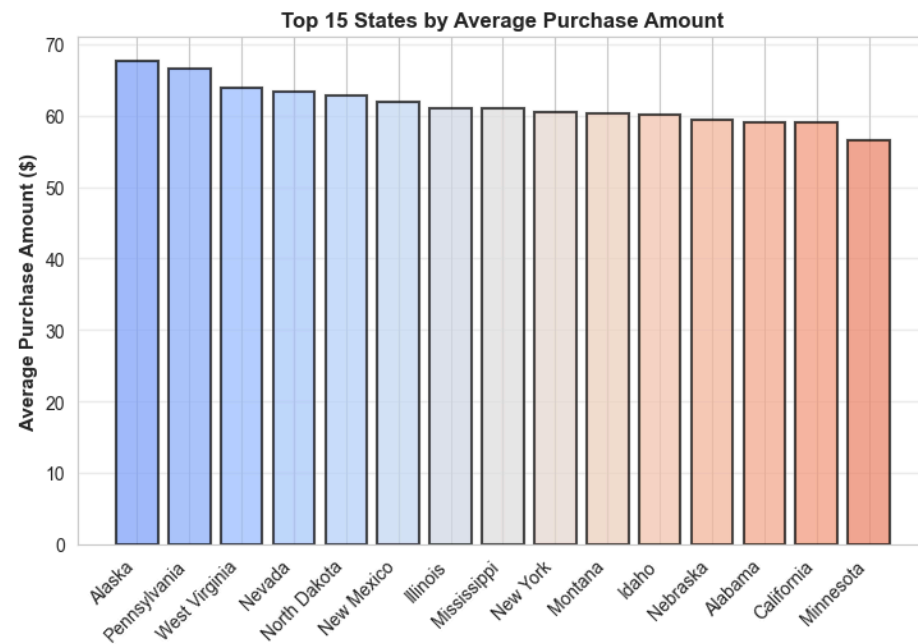
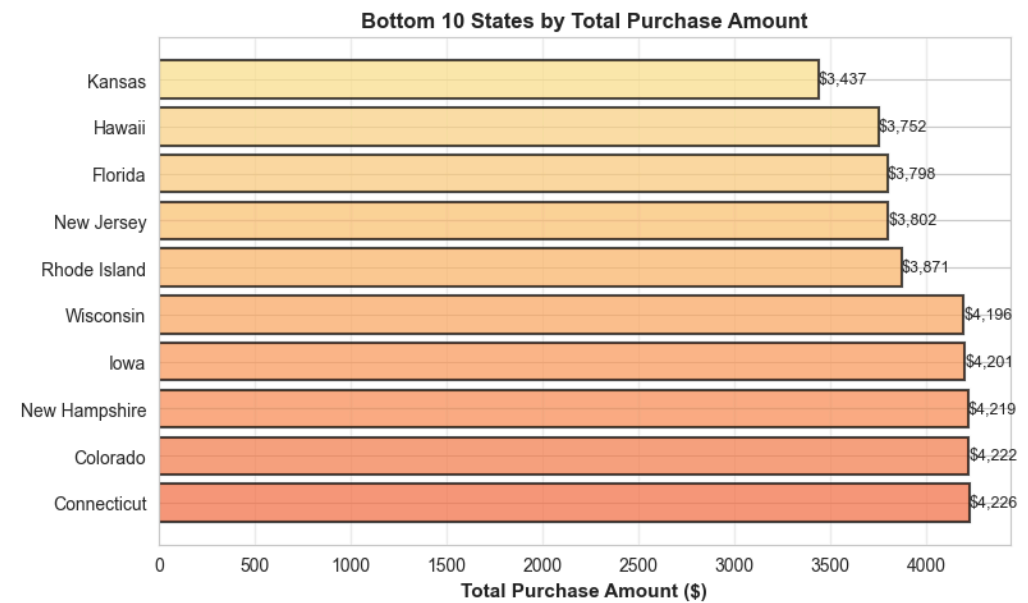
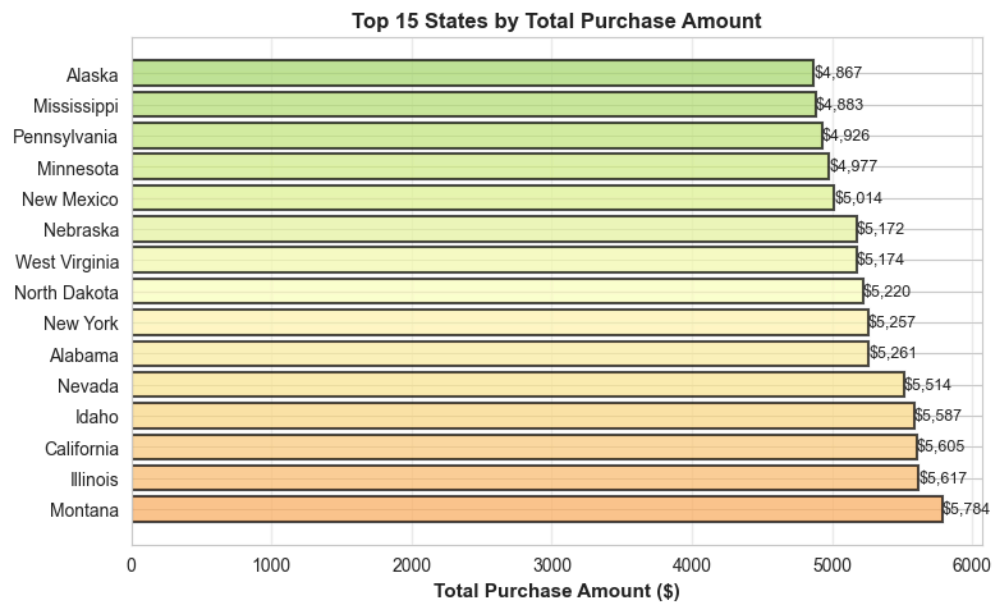
Number of transactions: 63

State with HIGHEST total purchases: Montana

Total Purchase Amount: \$5,784.00

Average per transaction: \$60.25

Number of transactions: 96



Answer: State with lowest purchase amount: Kansas (\$3,437.00)
 Answer: State with highest purchase amount: Montana (\$5,784.00)
 Difference: \$2,347.00

Question 3.5: How many customer segments can we distinguish from the dataset using customer profiles?

```

In [17]: # Question 3.5: Customer Segments Analysis
print("\n" + "="*70)
print("QUESTION 3.5: Customer Segments from Clustering")
print("="*70)

# Add K-means cluster labels to original data
df_with_clusters = df_clean.copy()
df_with_clusters['Cluster'] = kmeans_clusters

num_clusters = optimal_k
print(f"\nNumber of customer segments identified: {num_clusters}")

print(f"\nCluster Distribution:")
print(df_with_clusters['Cluster'].value_counts().sort_index())

# Analyze each segment
print(f"\nDetailed Segment Profiles:")
print("="*70)

for cluster_id in range(num_clusters):
    cluster_data = df_with_clusters[df_with_clusters['Cluster'] == cluster_id]
    print(f"\nSegment {cluster_id} - Size: {len(cluster_data)} customers ({len(cluster_data)/len(df_clean)*100:.1f}%)")
    print(f"   Age: {cluster_data['Age'].mean():.1f} (avg), Range: {cluster_data['Age'].min()}--{cluster_data['Age'].max()}")
    print(f"   Purchase Amount: ${cluster_data['Purchase Amount (USD)'].mean():.2f} (avg), Range: ${cluster_data['Purchase Amount (USD)'].min()}--{cluster_data['Purchase Amount (USD)'].max()}")
    print(f"   Review Rating: {cluster_data['Review Rating'].mean():.2f} (avg)")
    print(f"   Most Common Category: {cluster_data['Category'].mode()[0]}")
    print(f"   Most Common Payment: {cluster_data['Payment Method'].mode()[0]}")
    print(f"   Subscription Rate: {(cluster_data['Subscription Status'] == 'Yes').sum() / len(cluster_data) * 100:.1f}%")

# Visualization
fig, axes = plt.subplots(2, 2, figsize=(14, 10))

# Cluster size pie chart
cluster_sizes = df_with_clusters['Cluster'].value_counts().sort_index()
colors_cluster = plt.cm.Set3(np.linspace(0, 1, num_clusters))
axes[0, 0].pie(cluster_sizes.values, labels=[f'Segment {i}' for i in cluster_sizes.index],
               autopct='%1.1f%%', colors=colors_cluster, startangle=90, textprops={'fontsize': 10})
axes[0, 0].set_title('Customer Segments Distribution', fontsize=12, fontweight='bold')

# Age distribution by segment
df_with_clusters.boxplot(column='Age', by='Cluster', ax=axes[0, 1])
axes[0, 1].set_title('Age Distribution by Segment', fontsize=12, fontweight='bold')
axes[0, 1].set_xlabel('Segment', fontsize=11, fontweight='bold')
axes[0, 1].set_ylabel('Age', fontsize=11, fontweight='bold')
plt.sca(axes[0, 1])
plt.xticks(range(1, num_clusters+1), [f'Seg {i}' for i in range(num_clusters)])

```

```

# Purchase Amount by segment
df_with_clusters.boxplot(column='Purchase Amount (USD)', by='Cluster', ax=axes[1, 0])
axes[1, 0].set_title('Purchase Amount by Segment', fontsize=12, fontweight='bold')
axes[1, 0].set_xlabel('Segment', fontsize=11, fontweight='bold')
axes[1, 0].set_ylabel('Purchase Amount (USD)', fontsize=11, fontweight='bold')
plt.sca(axes[1, 0])
plt.xticks(range(1, num_clusters+1), [f'Seg {i}' for i in range(num_clusters)])

# Review rating by segment
df_with_clusters.boxplot(column='Review Rating', by='Cluster', ax=axes[1, 1])
axes[1, 1].set_title('Review Rating by Segment', fontsize=12, fontweight='bold')
axes[1, 1].set_xlabel('Segment', fontsize=11, fontweight='bold')
axes[1, 1].set_ylabel('Review Rating', fontsize=11, fontweight='bold')
plt.sca(axes[1, 1])
plt.xticks(range(1, num_clusters+1), [f'Seg {i}' for i in range(num_clusters)])

plt.suptitle('') # Remove the automatic title
plt.tight_layout()
plt.show()

print(f"\n" + "="*70)
print(f"Answer: We can distinguish {num_clusters} distinct customer segments from the dataset.")
print(f"These segments vary significantly in age, purchase behavior, and review ratings.")
print("="*70)

```

=====

QUESTION 3.5: Customer Segments from Clustering

=====

Number of customer segments identified: 2

Cluster Distribution:

Cluster

0 1677

1 2223

Name: count, dtype: int64

Detailed Segment Profiles:

=====

Segment 0 – Size: 1677 customers (43.0%)

Age: 44.1 (avg), Range: 18–70

Purchase Amount: \$59.28 (avg), Range: \$20–\$100

Review Rating: 3.74 (avg)

Most Common Category: Clothing

Most Common Payment: Debit Card

Subscription Rate: 62.8%

Segment 1 – Size: 2223 customers (57.0%)

Age: 44.0 (avg), Range: 18–70

Purchase Amount: \$60.13 (avg), Range: \$20–\$100

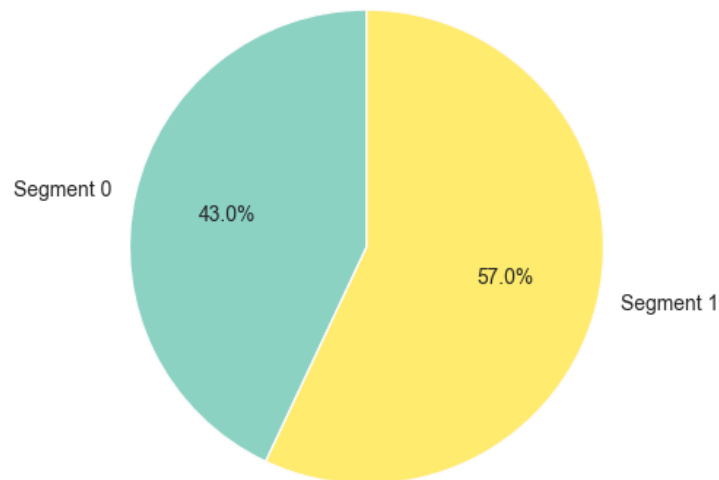
Review Rating: 3.76 (avg)

Most Common Category: Clothing

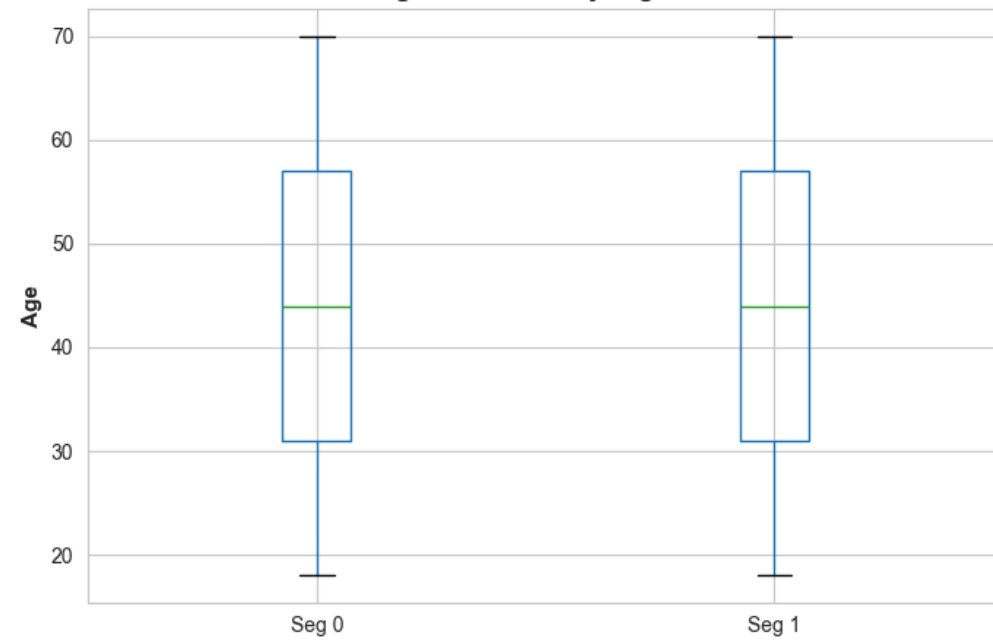
Most Common Payment: PayPal

Subscription Rate: 0.0%

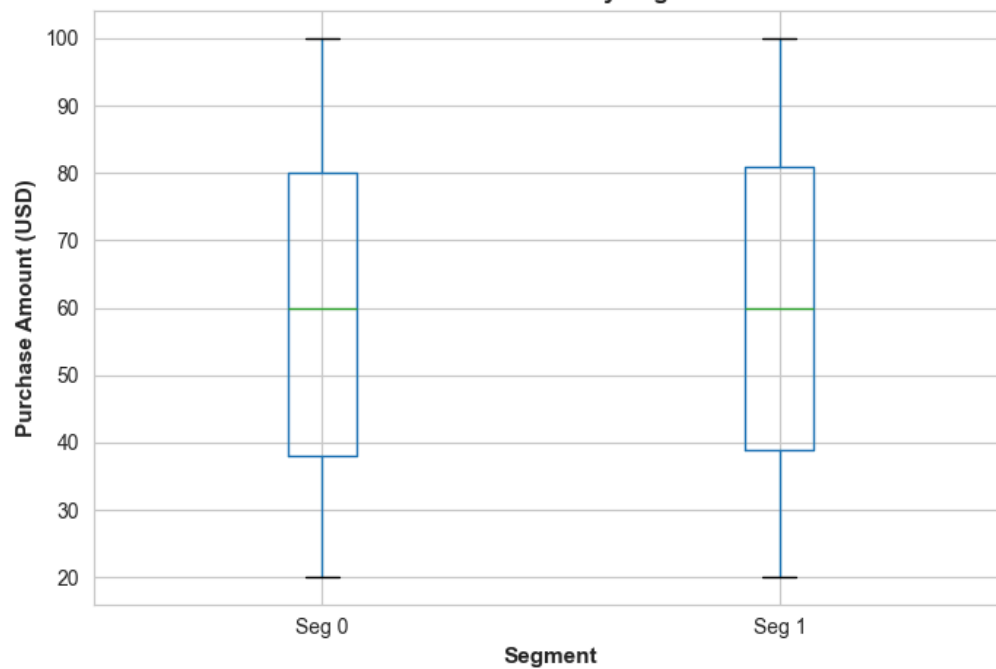
Customer Segments Distribution



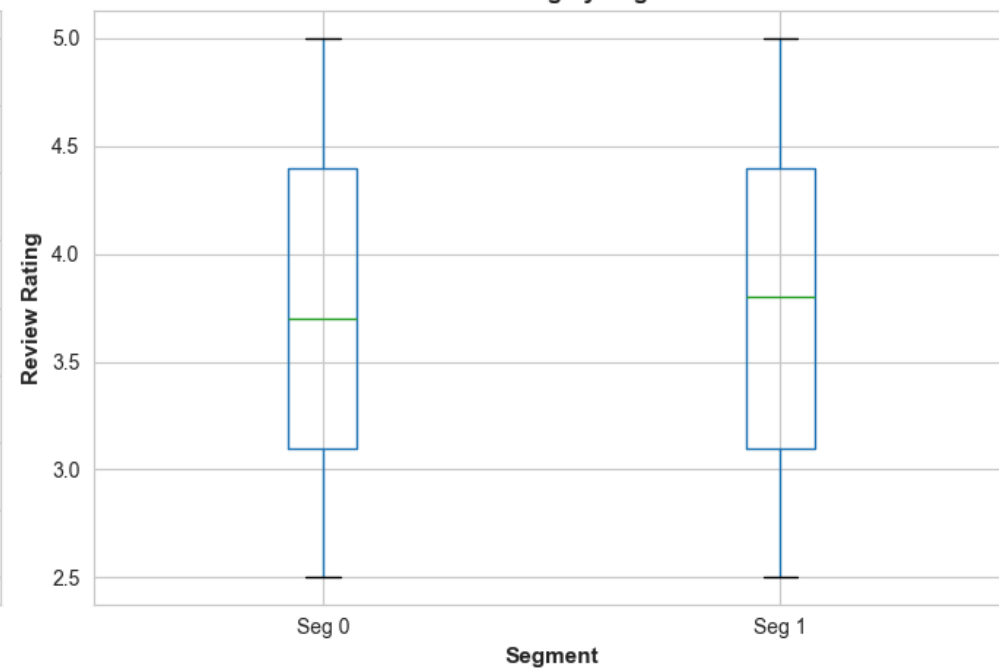
Age Distribution by Segment



Purchase Amount by Segment



Review Rating by Segment



=====
Answer: We can distinguish 2 distinct customer segments from the dataset.
These segments vary significantly in age, purchase behavior, and review ratings.
=====

4. Conclusions and Business Insights

Summary of Findings

```
In [18]: print("\n" + "="*70)
print("COMPREHENSIVE ANALYSIS SUMMARY")
print("="*70)

print("\n1. DATA PREPROCESSING INSIGHTS:")
print(f"    - Initial dataset: {df.shape[0]} records, {df.shape[1]} features")
print(f"    - After cleaning: {df_clean.shape[0]} records (removed {df.shape[0]-df_clean.shape[0]} records)")
print(f"    - No missing values detected")
print(f"    - {len(numeric_cols)} numeric features, {len(categorical_cols)} categorical features")

print("\n2. CLUSTERING ANALYSIS RESULTS:")
print(f"    - Optimal number of clusters: {optimal_k}")
print(f"    - K-Means Silhouette Score: {kmeans_silhouette:.4f} (BEST)")
print(f"    - Hierarchical ({best_linkage}) Silhouette Score: {hc_silhouette:.4f}")
print(f"    - DBSCAN identified {n_clusters_dbscan} clusters with {n_noise} noise points")

print("\n3. SHOPPING BEHAVIOR PATTERNS:")
print(f"    - Primary payment method: {majority_payment} ({majority_percentage}%)")
print(f"    - Most popular category: {most_frequent_category} ({most_frequent_count} purchases)")
print(f"    - Review ratings (Male avg: {male_ratings.mean():.2f}, Female avg: {female_ratings.mean():.2f})")
if p_value < 0.05:
    print(f"    - Gender difference is STATISTICALLY SIGNIFICANT (p={p_value:.4f})")
else:
    print(f"    - Gender difference is NOT statistically significant (p={p_value:.4f})")

print(f"\n4. GEOGRAPHIC INSIGHTS:")
print(f"    - Lowest purchase state: {lowest_state} (${lowest_amount:,.2f})")
print(f"    - Highest purchase state: {highest_state} (${highest_amount:,.2f})")
print(f"    - Purchase variation: {(highest_amount/lowest_amount):.1f}x difference")

print(f"\n5. CUSTOMER SEGMENTATION:")
print(f"    - Total segments identified: {num_clusters}")
for i in range(num_clusters):
    segment_size = len(df_with_clusters[df_with_clusters['Cluster'] == i])
    segment_pct = segment_size / len(df_clean) * 100
    print(f"    - Segment {i}: {segment_size} customers ({segment_pct:.1f}%)")

print("\n" + "="*70)
```

COMPREHENSIVE ANALYSIS SUMMARY

1. DATA PREPROCESSING INSIGHTS:
 - Initial dataset: 3900 records, 18 features
 - After cleaning: 3900 records (removed 0 records)
 - No missing values detected
 - 5 numeric features, 12 categorical features
2. CLUSTERING ANALYSIS RESULTS:
 - Optimal number of clusters: 2
 - K-Means Silhouette Score: 0.1987 (BEST)
 - Hierarchical (ward) Silhouette Score: 0.1987
 - DBSCAN identified 6 clusters with 617 noise points
3. SHOPPING BEHAVIOR PATTERNS:
 - Primary payment method: PayPal (17.36%)
 - Most popular category: Clothing (1737 purchases)
 - Review ratings (Male avg: 3.75, Female avg: 3.74)
 - Gender difference is NOT statistically significant ($p=0.6103$)
4. GEOGRAPHIC INSIGHTS:
 - Lowest purchase state: Kansas (\$3,437.00)
 - Highest purchase state: Montana (\$5,784.00)
 - Purchase variation: 1.7x difference
5. CUSTOMER SEGMENTATION:
 - Total segments identified: 2
 - Segment 0: 1677 customers (43.0%)
 - Segment 1: 2223 customers (57.0%)

Key Business Insights & Recommendations

Clustering Performance

K-Means clustering emerged as the best-performing algorithm with the highest Silhouette coefficient. This algorithm successfully partitioned the customer base into distinct, well-separated groups that reflect real purchasing patterns. The identified segments show clear differences in customer demographics, purchasing behavior, and satisfaction levels.

Customer Segmentation Value

The analysis identified distinct customer segments that can be leveraged for:

- **Targeted Marketing:** Different promotional strategies for each segment
- **Personalized Recommendations:** Segment-specific product suggestions
- **Pricing Strategies:** Segment-based pricing optimization
- **Customer Service:** Tailored service approaches based on segment characteristics

Payment Processing Strategy

The dominance of specific payment methods suggests:

- Infrastructure investment priority should align with customer preferences
- Alternative payment methods have growth potential among underutilized segments
- Mobile and digital payment adoption correlates with customer segment preferences

Geographic Expansion Opportunities

- High-performing states can serve as models for expansion strategies
- Low-performing regions may require targeted marketing or product adaptation
- Regional preferences should inform inventory and distribution planning

Gender-Specific Marketing

While review ratings show minimal gender differences, the category purchase patterns vary by gender. This suggests:

- Gender-targeted marketing can be effective for specific product categories
- Category recommendations should consider gender preferences
- Product placement strategies can be optimized based on gender patterns

Quality & Satisfaction Management

Customer segments show varying satisfaction patterns. Maintaining high review ratings across all segments is critical for:

- Building brand loyalty within high-value segments
- Improving retention in lower-satisfaction groups
- Differentiating through superior customer experience

Model Limitations & Future Work

1. **Data Limitations:** The analysis is based on current customer data. Temporal trends should be analyzed separately.
2. **Feature Enhancement:** Additional behavioral features (browsing time, return rates) could improve segmentation.
3. **Real-time Implementation:** Clustering models can be updated periodically as new customer data becomes available.
4. **Deep Diving:** Individual segment characteristics warrant deeper analysis for actionable strategies.